Unit 1: Numerical Expressions & Factors — Complete Key

Big Ideas Math Grade 6 — Chapter 1 • Manipulative Kits: Counting & Array Studio (COUNT-01) + Prime Factor Studio (PRIME-01)

Target Misconception: Students interpret remainders as meaningless decimal tails (e.g., .875 of a person), read 2^3 as 2×3=6, lose track of PEMDAS parentheses, and confuse GCF (overlap) with LCM (union).

Hands-On Block Remediation: Have students build physical tile rows for division remainders, stack 2D/3D arrays for exponents, use the 4-line elevator (Y holds Room 1 while X solves Room 2) instead of rewriting PEMDAS lines, and stand logarithmic Prime Factor Towers side-by-side on the 12-socket lid.



Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!

SEC 1.1: WHOLE NUMBER OPERATIONS & DIVISION WITH REMAINDERS [COUNT-01]

Method: Rectangular Array Model & Division Check (Dividend = Divisor × Quotie..

#1: Arrange 31 tiles into rows of 6. Find Quotient & Remainder.
SC-6 Keys: `31 [ENTER] 6 [g] [÷R]` **Ans: 5 R 1**

#2: Divide 47 ÷ 5 using tiles or [g] [÷R].
SC-6 Keys: `47 [ENTER] 5 [g] [÷R]` **Ans: 9 R 2**

#3: Divide 58 ÷ 7 and check with [f] [LASTX] [×] [+].
SC-6 Keys: `58 [ENTER] 7 [g] [÷R]` **Ans: 8 R 2**

#4: Divide 85 ÷ 9. How many full groups of 9 and how many left?
SC-6 Keys: `85 [ENTER] 9 [g] [÷R]` **Ans: 9 R 4**

#5: Divide 146 ÷ 12. Find the Quotient and Remainder.
SC-6 Keys: `146 [ENTER] 12 [g] [÷R]` **Ans: 12 R 2**

#6: Divide 250 ÷ 36 and prove with [f] [LASTX] [×] [+].
SC-6 Keys: `250 [ENTER] 36 [g] [÷R]` **Ans: 6 R 34**

#7: Find the missing dividend: Quotient = 14, Divisor = 8, Remainder = 5.
SC-6 Keys: `14 [ENTER] 8 [×] 5 [+]` **Ans: 117**

#8: Spot the Glitch: Sam says 75 ÷ 8 = 8 R 11. Why is R=11 impossible? Fix it!
SC-6 Keys: `75 [ENTER] 8 [g] [÷R]` **Ans: 9 R 3 (Rem < 8)**

#9*: A 3D spool has 300g of PLA. Each box takes 42g. How many full boxes can you print, and how much PLA i..
SC-6 Keys: `300 [ENTER] 42 [g] [÷R]` **Ans: 7 boxes, 6g left (Way 2: 7..**

#10*: 115 students ride vans that hold 14 kids each. How many vans must be booked so nobody is left behind?
SC-6 Keys: `115 [ENTER] 14 [g] [÷R]` **Ans: 9 vans (8 full + 1 for 3 k..**

SEC 1.2: POWERS & EXPONENTS — SQUARES, CUBES & GROWTH [COUNT-01]

Method: Repeated Multiplication Factors, Perfect Squares (2D Array) & Cubes (..

#1: Write 5^2 as repeated multiplication and find how many tiles are in a 5 × 5 square.
SC-6 Keys: `5 [x²]` **Ans: 25**

#2: Evaluate 7^2 and compare it to 7 × 2.
SC-6 Keys: `7 [x²]` **Ans: 49 (vs 14)**

#3: Evaluate 3^3 (a 3 × 3 × 3 cube of blocks).
SC-6 Keys: `3 [g] [x³]` **Ans: 27**

#4: Evaluate 5^3 using repeated multiplication or [g] [x^3].
SC-6 Keys: `5 [g] [x³]` **Ans: 125**

#5: Evaluate 2^5 by writing the repeated factors and checking with 2 [ENTER] 5 [g] [y^x].
SC-6 Keys: `2 [ENTER] 5 [g] [y^x]` **Ans: 32**

#6: Evaluate 6^4 using repeated multiplication or [g] [y^x].
SC-6 Keys: `6 [ENTER] 4 [g] [y^x]` **Ans: 1,296**

#7: Evaluate 10^4 and 1^4. How many zeros are in 10^4?
SC-6 Keys: `10 [ENTER] 4 [g] [y^x]` **Ans: 10,000 and 1**

#8: Spot the Glitch: Mia wrote 2^4 = 8 because 2 × 4 = 8. Find the true value of 2^4.
SC-6 Keys: `2 [ENTER] 4 [g] [y^x]` **Ans: 16 (2×2×2×2)**

#9*: A cell doubles every hour (2^n). How many cells are there after 6 hours (2^6)? Solve 2 ways!
SC-6 Keys: `2 [ENTER] 6 [g] [y^x]` **Ans: 64 cells (Way 2: 2 [g] [REP..**

#10*: Mystery Cube: A giant cube uses 216 unit blocks. How many blocks wide is 1 edge? (Try [g] [x√y]!)
SC-6 Keys: `216 [ENTER] 3 [g] [x√y]` **Ans: 6 blocks wide (Way 2: 6 [g..**

SEC 1.3: ORDER OF OPERATIONS — GROUPING SYMBOLS, EXPONENTS & PEMDAS [COUNT-01]

Method: Order of Operations (PEMDAS: Grouping Symbols -> Powers -> Multiply/D..

#1: Evaluate 5 + 2 × 3. Multiply before adding!
SC-6 Keys: `5 [ENTER] 2 [ENTER] 3 [×] [+]` **Ans: 11**

#2: Evaluate (14 + 6) ÷ (7 - 2). Solve both parentheses first!
SC-6 Keys: `14 [ENTER] 6 [+] 7 [ENTER] 2 [-] [÷]` **Ans: 4**

#3: Evaluate 30 - 4 × 6.
SC-6 Keys: `30 [ENTER] 4 [ENTER] 6 [×] [-]` **Ans: 6**

#4: Evaluate (9 + 3) × (15 - 8).
SC-6 Keys: `9 [ENTER] 3 [+] 15 [ENTER] 8 [-] [×]` **Ans: 84**

#5: Evaluate 50 - (3 + 4)². Square the sum before subtracting from 50!
SC-6 Keys: `50 [ENTER] 3 [ENTER] 4 [+] [x²] [-]` **Ans: 1**

#6: Evaluate 120 - 4 × (3 + 2)².
SC-6 Keys: `120 [ENTER] 4 [ENTER] 3 [ENTER] 2 [+]..` **Ans: 20**

#7: Fraction Bar Grouping: Evaluate (24 + 12) / (9 - 3).
SC-6 Keys: `24 [ENTER] 12 [+] 9 [ENTER] 3 [-] [÷]` **Ans: 6**

#8: Evaluate 3³ + 15 ÷ 3 - 2 × 4.
SC-6 Keys: `3 [g] [x³] 15 [ENTER] 3 [÷] [+] 2 [ENT..` **Ans: 24**

#9*: Evaluate 10 + (15 - 3)² ÷ 9 using PEMDAS step-by-step AND check on your SC-6.
SC-6 Keys: `10 [ENTER] 15 [ENTER] 3 [-] [x²] 9 [÷..` **Ans: 26 (Way 2: 10 + 144÷9 = 26)**

#10*: Movie tickets: 2 adults at $12 each and 4 kids at $8 each, minus a $10 coupon. Write an expression and fin..
SC-6 Keys: `2 [ENTER] 12 [×] 4 [ENTER] 8 [×] [+]..` **Ans: $46 (Way 2: 24 + 32 - 10 =..**

SEC 1.4: PRIME FACTORIZATION — FACTOR TREES & LOGARITHMIC PRIME TOWERS [PRIME-01]

Method: Factor Trees, Prime vs. Composite Testing & Prime-Power Factorization..

#1: Draw a factor tree or stack [2] + [2] + [3]. Write the prime factorization of 12.
SC-6 Keys: `12 [g] [FACT]` **Ans: $2^2 \times 3$**

#2: Find the prime factorization of 36 using blocks and [g] [FACT].
SC-6 Keys: `36 [g] [FACT]` **Ans: $2^2 \times 3^2$**

#3: Test 29 and 57 with [PRIME]. Which one is prime and which is composite?
SC-6 Keys: `29 [PRIME], 57 [g] [FACT]` **Ans: 29 Prime; 57 Composite (3×..**

#4: Find the prime factorization of 45.
SC-6 Keys: `45 [g] [FACT]` **Ans: $3^2 \times 5$**

#5: Build the Prime Tower for 72. Show how 8 × 9 and 6 × 12 use the exact same blocks.
SC-6 Keys: `72 [g] [FACT]` **Ans: $2^3 \times 3^2$**

#6: Find the prime factorization of 84.
SC-6 Keys: `84 [g] [FACT]` **Ans: $2^2 \times 3 \times 7$**

#7: Find the prime factorization of 100.
SC-6 Keys: `100 [g] [FACT]` **Ans: $2^2 \times 5^2$**

#8: What number has the prime tower [2] + [2] + [3] + [7] ($2^2 \times 3 \times 7$)?
SC-6 Keys: `2 [x²] 3 [×] 7 [×]` **Ans: 84**

#9*: Find the prime factorization of 180 both by splitting 18 × 10 AND with [g] [FACT].
SC-6 Keys: `180 [g] [FACT]` **Ans: $2^2 \times 3^2 \times 5$ (Way 2: (2·3²)..**

#10*: Mystery Vault Code: A 3-digit number equals $2^3 \times 3^2 \times 5$. Find the number and verify with [g] [FACT].
SC-6 Keys: `2 [g] [x³] 3 [x²] [×] 5 [×]` **Ans: 360 (Way 2: 8 × 9 × 5 = 360)**

SEC 1.5: GREATEST COMMON FACTOR (GCF) & LEAST COMMON MULTIPLE (LCM) [PRIME-01]

Method: Venn Diagram of Prime Factors — Shared Overlap for GCF & Tallest Prim..

#1: Find the GCF and LCM of 12 and 18 using Prime Towers and [GCF] / [f] [LCM].
SC-6 Keys: `12 [ENTER] 18 [GCF] / [f] [LCM]` **Ans: GCF = 6, LCM = 36**

#2: Find the GCF and LCM of 8 and 12.
SC-6 Keys: `8 [ENTER] 12 [GCF] / [f] [LCM]` **Ans: GCF = 4, LCM = 24**

#3: Find the GCF and LCM of 15 and 25.
SC-6 Keys: `15 [ENTER] 25 [GCF] / [f] [LCM]` **Ans: GCF = 5, LCM = 75**

#4: Find the GCF and LCM of 14 and 35.
SC-6 Keys: `14 [ENTER] 35 [GCF] / [f] [LCM]` **Ans: GCF = 7, LCM = 70**

#5: Find the GCF and LCM of 16 and 24.
SC-6 Keys: `16 [ENTER] 24 [GCF] / [f] [LCM]` **Ans: GCF = 8, LCM = 48**

#6: Find the GCF and LCM of 28 and 42.
SC-6 Keys: `28 [ENTER] 42 [GCF] / [f] [LCM]` **Ans: GCF = 14, LCM = 84**

#7: Two prime numbers are 5 and 7. What is their GCF and what is their LCM?
SC-6 Keys: `5 [ENTER] 7 [GCF] / [f] [LCM]` **Ans: GCF = 1, LCM = 35**

#8: Verify that GCF(15, 20) × LCM(15, 20) equals 15 × 20.
SC-6 Keys: `15 [ENTER] 20 [GCF] ... [×]` **Ans: 5 × 60 = 300 (Matches 15×2..**

#9*: Hot dogs come in packs of 12 and buns in packs of 15. What is the smallest number of each to buy with 0 le..
SC-6 Keys: `12 [ENTER] 15 [f] [LCM]` **Ans: 60 (5 packs dogs, 4 packs..**

#10*: You have 36 red tiles and 48 blue tiles to pack into identical kit bags with none left over. What is the max nu..
SC-6 Keys: `36 [ENTER] 48 [GCF]` **Ans: 12 bags (3 red, 4 blue per..**

Unit 2: Fractions & Decimals — Complete Key

Big Ideas Math Grade 6 — Chapter 2 • Manipulative Kits: Fraction Studio (FRAC-01) + Place Value Studio (PLACE-01)

Target Misconception: Students treat fractions, mixed numbers, and decimals as unrelated rules, expect multiplication to always make numbers bigger, forget which fraction to flip in division, and misalign decimal columns.

Hands-On Block Remediation: Use the 84.0 mm Fraction Studio towers so equivalence and common denominators are physical millimeter height matches. Use [f] [1/x] on Floor 1 (X) so students see only the divisor flip, and use [f] [IP] / [g] [FP] to split whole dollars from decimal cents.

Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!



SEC 2.1: EQUIVALENT FRACTIONS, MIXED NUMBERS & ADDING TOWERS [FRAC-01]

Method: Equivalent Fractions via GCF Simplification & Adding/Subtracting with..

#1: Simplify 8/12 using [g] [SIMP]. What intermediate fraction appears first?
SC-6 Keys: [.] 8 [.] 12 [ENTER] [g][SIMP]×2 — **Ans: 4/6 -> 2/3**

#2: Simplify 24/36 completely and check its decimal with [FDISP].
SC-6 Keys: [.] 24 [.] 36 [ENTER] [g][SIMP]... — **Ans: 2/3 (0.667)**

#3: Convert 2 3/5 into an improper fraction ([f] [d/c]) and a decimal ([FDISP]).
SC-6 Keys: 2 [.] 3 [.] 5 [ENTER] [f][d/c] — **Ans: 13/5 and 2.6**

#4: Convert 19/8 into a mixed number ([a b/c]) and a decimal ([FDISP]).
SC-6 Keys: [.] 19 [.] 8 [ENTER] [FDISP] — **Ans: 2 3/8 and 2.375**

#5: Add 1/2 + 1/3 using a common denominator (sixths) and check with 84mm Fraction Blocks.
SC-6 Keys: [.] 1 [.] 2 [ENTER] [.] 1 [.] 3 [+] — **Ans: 5/6 (42mm+28mm=70mm)**

#6: Add 3/4 + 5/6 using the LCD (12). Write both the mixed number and improper fraction.
SC-6 Keys: [.] 3 [.] 4 [ENTER] [.] 5 [.] 6 [+] — **Ans: 1 7/12 (or 19/12)**

#7: Subtract 7/8 - 1/4 and verify with [f] [LASTX] [+].
SC-6 Keys: [.] 7 [.] 8 [ENTER] [.] 1 [.] 4 [-] — **Ans: 5/8**

#8: Subtract mixed numbers: 4 1/3 - 1 5/6.
SC-6 Keys: 4 [.] 1 [.] 3 [ENTER] 1 [.] 5 [.] 6 [-] — **Ans: 2 1/2 (or 5/2)**

#9*: Build 1/3 + 1/4 with 84mm towers (28mm + 21mm) AND solve using the LCD (twelfths).
SC-6 Keys: [.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [+] — **Ans: 7/12 (49mm = seven 7mm twe..**

#10*: A trail is 3 3/4 miles long. You hiked 1 7/8 miles. How much trail is left? Solve 2 ways!
SC-6 Keys: 3 [.] 3 [.] 4 [ENTER] 1 [.] 7 [.] 8 [-] — **Ans: 1 7/8 miles (Way 2: 30/8 -..**

SEC 2.2: MULTIPLYING FRACTIONS & MIXED NUMBERS [FRAC-01]

Method: Area/Tower Model for Fraction × Fraction & Cross-Canceling Common Fac..

#1: Find 1/2 × 2/3. (Hold two 1/3 blocks and take half of them!)
SC-6 Keys: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×] — **Ans: 1/3**

#2: Find 2/3 × 3/4 and check with [f] [LASTX] [÷].
SC-6 Keys: [.] 2 [.] 3 [ENTER] [.] 3 [.] 4 [×] — **Ans: 1/2**

#3: Multiply 3/5 × 5/9.
SC-6 Keys: [.] 3 [.] 5 [ENTER] [.] 5 [.] 9 [×] — **Ans: 1/3**

#4: Multiply a whole number by a fraction: 12 × 3/4.
SC-6 Keys: 12 [ENTER] [.] 3 [.] 4 [×] — **Ans: 9**

#5: Multiply mixed numbers: 1 3/4 × 2 2/7.
SC-6 Keys: 1 [.] 3 [.] 4 [ENTER] 2 [.] 2 [.] 7 [×] — **Ans: 4**

#6: Multiply 2 1/4 × 1 1/3.
SC-6 Keys: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×] — **Ans: 3**

#7: Multiply 3 1/2 × 2 1/2. Write the answer as a mixed number and decimal.
SC-6 Keys: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 2 [×] — **Ans: 8 3/4 (8.75)**

#8: Prediction Check: Without calculating, is 5 × 2/3 greater than 5 or less than 5? Now compute it!
SC-6 Keys: 5 [ENTER] [.] 2 [.] 3 [×] — **Ans: 3 1/3 (Less than 5!)**

#9*: A slime recipe uses 1 3/4 cups of glue per batch. How much glue is needed for 2 1/2 batches?
SC-6 Keys: 1 [.] 3 [.] 4 [ENTER] 2 [.] 1 [.] 2 [×] — **Ans: 4 3/8 cups (Way 2: 1.75 ×..**

#10*: Triple Chain: Evaluate 4/5 × 5/6 × 3/4 both by cross-canceling common factors AND on your SC-6!
SC-6 Keys: [.] 4 [.] 5 [ENTER] [.] 5 [.] 6 [×] [— **Ans: 1/2 (Way 2: cancel 4s & 5s..**

SEC 2.3: DIVIDING FRACTIONS & MIXED NUMBERS — WHY RECIPROCAL FLIP WORKS [FRAC-01]

Method: Measurement Division ('How Many Fit?') & Multiplying by the Reciproca..

#1: How many 1/4 blocks fit in a 1/2 tower? Compute 1/2 ÷ 1/4.
SC-6 Keys: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [÷] — **Ans: 2**

#2: How many 1/6 blocks (14mm) fit in a 2/3 tower (56mm)? Compute 2/3 ÷ 1/6.
SC-6 Keys: [.] 2 [.] 3 [ENTER] [.] 1 [.] 6 [÷] — **Ans: 4**

#3: Divide 5/6 ÷ 5/12 by multiplying by the reciprocal ([f] [1/x] [×]).
SC-6 Keys: [.] 5 [.] 6 [ENTER] [.] 5 [.] 12 [f][.. — **Ans: 2**

#4: Divide a whole number by a fraction: 6 ÷ 3/4.
SC-6 Keys: 6 [ENTER] [.] 3 [.] 4 [÷] — **Ans: 8**

#5: Divide 3/4 ÷ 2/5. Write the answer as a mixed number.
SC-6 Keys: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [÷] — **Ans: 1 7/8 (15/8)**

#6: Divide a fraction by a whole number: 4/5 ÷ 2.
SC-6 Keys: [.] 4 [.] 5 [ENTER] 2 [÷] — **Ans: 2/5**

#7: Divide mixed numbers: 4 1/2 ÷ 3/4.
SC-6 Keys: 4 [.] 1 [.] 2 [ENTER] [.] 3 [.] 4 [÷] — **Ans: 6**

#8: Divide 5 1/4 ÷ 1 3/4 and prove with [f] [LASTX] [×].
SC-6 Keys: 5 [.] 1 [.] 4 [ENTER] 1 [.] 3 [.] 4 [÷] — **Ans: 3**

#9*: A board is 6 3/4 ft long. You cut it into 3/4-ft shelves. How many shelves can you cut? Solve 2 ways!
SC-6 Keys: 6 [.] 3 [.] 4 [ENTER] [.] 3 [.] 4 [÷] — **Ans: 9 shelves (Way 2: 27 fourt..**

#10*: A 3 3/4-cup pitcher of smoothie is poured into 5/8-cup glasses. How many glasses does it fill?
SC-6 Keys: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷] — **Ans: 6 glasses (Way 2: 30/8 ÷ 5..**

SEC 2.4: ADDING, SUBTRACTING & MULTIPLYING DECIMALS [PLACE-01]

Method: Vertical Decimal Point Alignment (+/-) & Counting Decimal Places / Es..

#1: Enter 27.64. Use [f] [IP] and [g] [FP] to write its Whole part and Decimal part.
SC-6 Keys: 27.64 [f][IP] / [g][FP] — **Ans: Whole = 27, Decimal = 0.64**

#2: Line up the decimal points to add 12.8 + 4.75, and check with [f] [LASTX] [-].
SC-6 Keys: 12.8 [ENTER] 4.75 [+] — **Ans: 17.55**

#3: Annex zeros to subtract 20.00 - 8.65.
SC-6 Keys: 20 [ENTER] 8.65 [-] — **Ans: 11.35**

#4: Multiply 4.5 × 0.6.
SC-6 Keys: 4.5 [ENTER] 0.6 [×] — **Ans: 2.7**

#5: Evaluate 25.4 - 9.86 + 3.2.
SC-6 Keys: 25.4 [ENTER] 9.86 [-] 3.2 [+] — **Ans: 18.74**

#6: Multiply 3.6 × 2.25 and check the whole-number ballpark.
SC-6 Keys: 3.6 [ENTER] 2.25 [×] — **Ans: 8.1**

#7: Multiply 0.15 × 0.4 and tap [FDISP] to see its simplified fraction.
SC-6 Keys: 0.15 [ENTER] 0.4 [×] [FDISP] — **Ans: 0.06 (3/50)**

#8: Find the area of a rectangular desk top measuring 1.8 m by 0.75 m.
SC-6 Keys: 1.8 [ENTER] 0.75 [×] — **Ans: 1.35 m²**

#9*: You pay with a $20 bill for a $8.75 burger and $4.60 smoothie. How much change do you get?
SC-6 Keys: 20 [ENTER] 8.75 [ENTER] 4.60 [+] [-] — **Ans: $6.65 (Way 2: 20 - 8.75 -..**

#10*: Multiply 4.75 × 1.2 both as decimals AND as fractions (4 3/4 × 1 1/5) using [FDISP]!
SC-6 Keys: 4.75 [ENTER] 1.2 [×] — **Ans: 5.7 or 5 7/10 (Way 2: 19/4..**

SEC 2.5: DIVIDING DECIMALS & THE 3-WAY FRACTION-DECIMAL-REMAINDER TRANSLATOR [FRAC-01]

Method: Multiplying Dividend & Divisor by a Power of 10 to Make a Whole-Numbe..

#1: Write 39 ÷ 8 as (a) Quotient + Remainder, (b) Decimal, and (c) Mixed Number.
SC-6 Keys: 39 [ENTER] 8 [÷] [FDISP] — **Ans: 4 R 7, 4.875, 4 7/8**

#2: Divide a decimal by a whole number: 18.4 ÷ 8.
SC-6 Keys: 18.4 [ENTER] 8 [÷] — **Ans: 2.3**

#3: How many quarters ($0.25) are in $11.25? Compute 11.25 ÷ 0.25.
SC-6 Keys: 11.25 [ENTER] 0.25 [÷] — **Ans: 45**

#4: Divide 7.2 ÷ 0.9 and check with [f] [LASTX] [×].
SC-6 Keys: 7.2 [ENTER] 0.9 [÷] — **Ans: 8**

#5: Divide 15.6 ÷ 0.12.
SC-6 Keys: 15.6 [ENTER] 0.12 [÷] — **Ans: 130**

#6: Divide 9.45 ÷ 1.5 and tap [FDISP] to write its mixed fraction form.
SC-6 Keys: 9.45 [ENTER] 1.5 [÷] [FDISP] — **Ans: 6.3 (6 3/10)**

#7: Divide 0.84 ÷ 0.07.
SC-6 Keys: 0.84 [ENTER] 0.07 [÷] — **Ans: 12**

#8: Spot the Glitch: Leo says 4.8 ÷ 0.2 = 0.24. Use [f] [LASTX] or estimation to fix it!
SC-6 Keys: 4.8 [ENTER] 0.2 [÷] — **Ans: 24 (0.2 fits 5 times per 1)**

#9*: A runner completes a 4.5-mile race in 0.75-mile laps. How many laps is the race? Solve 2 ways!
SC-6 Keys: 4.5 [ENTER] 0.75 [÷] — **Ans: 6 laps (Way 2: 4 1/2 ÷ 3/4..**

#10*: A 6-pack of sports drinks costs $7.50. What is the cost of 1 bottle, and how many bottles can $20 buy?
SC-6 Keys: 7.50 [ENTER] 6 [÷] -> 1.25; 20 [ENTER.. — **Ans: $1.25/bottle; 16 bottles**

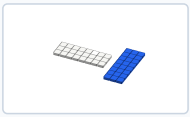
Unit 3: Ratios, Rates & Percents — Complete Key

Big Ideas Math Grade 6 — Chapters 3, 4 & 5 (Ratios, Rates & Percents) • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Fraction Studio (FRAC-01)

Target Misconception: Students confuse Part-to-Part (3:2) with Part-to-Whole (3/5), add instead of multiply in ratio tables, divide unit rates upside-down, and memorize 6 disconnected rules for moving decimal points in percents.

Hands-On Block Remediation: Use Blue and Gold Counting Tiles as physical Tape Diagram boxes, step through ratio tables rhythmically with [g] [REP], use [x↔y] when unit-rate division is entered upside-down, and use [f] [%] + [FDISP] to show Fraction/Decimal/Percent are three costumes of one number.

Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!



SEC 3.1: RATIOS & TAPE DIAGRAMS — PART-TO-PART VS. PART-TO-WHOLE [COUNT-01]

Method: Part-to-Part vs. Part-to-Whole Tape Diagrams & Equal-Share Box Value..

#1: Simplify the ratio 12 : 18 using [g] [a:b].
SC-6 Keys: `12 [ENTER] 18 [g] [a:b]` **Ans: 2 : 3**

#2: Simplify the ratio 24 : 16 using [g] [a:b].
SC-6 Keys: `24 [ENTER] 16 [g] [a:b]` **Ans: 3 : 2**

#3: A bag has 4 red marbles and 6 blue marbles. Find the simplified ratio of red to TOTAL marbles.
SC-6 Keys: `4 [ENTER] 10 [g] [a:b]` **Ans: 2 : 5**

#4: In a 3:4 tape diagram, each tile is worth 6. Find the two parts and the total.
SC-6 Keys: `3 [ENTER] 6 [×], 4 [ENTER] 6 [×]` **Ans: 18, 24 (Total = 42)**

#5: Green paint mixes blue and yellow in a 2:3 ratio. If you use 25 cups total, how many cups of each?
SC-6 Keys: `25 [ENTER] 5 [÷] -> 5/tile` **Ans: 10 blue, 15 yellow**

#6: The ratio of cats to dogs at a shelter is 4:5. There are 36 animals total. How many are dogs?
SC-6 Keys: `36 [ENTER] 9 [÷] 5 [×]` **Ans: 20 dogs**

#7: The ratio of boys to girls in a club is 3:5. If there are 15 boys, how many girls are there?
SC-6 Keys: `15 [ENTER] 3 [÷] 5 [×]` **Ans: 25 girls**

#8: Difference Tape Model: The ratio of acoustic to electric guitars is 2:5. There are 18 MORE electric guitars. F..
SC-6 Keys: `18 [ENTER] 3 [÷] 5 [×]` **Ans: 30 electric (12 acoustic)**

#9*: Two robot teams split 84 gears in a 4:3 ratio. How many gears does each team get? Solve 2 ways!
SC-6 Keys: `84 [ENTER] 7 [÷] -> 12/tile` **Ans: 48 and 36 gears (Way 2: 4/..**

#10*: Ratio of gold to silver coins is 7:4. There are 21 more gold coins than silver. How many coins in all?
SC-6 Keys: `21 [ENTER] 3 [÷] 11 [×]` **Ans: 77 coins total (Way 2: 49..**

SEC 3.2: RATIO TABLES & DOUBLE NUMBER LINES VIA [G] [REP] [COUNT-01]

Method: Equivalent Ratio Tables — Multiplying or Dividing Both Quantities by..

#1: Use 4 [g] [REP] and 9 [g] [REP] to write the first 4 equivalent ratios of 4 : 9.
SC-6 Keys: `4 [g] [REP] [+], 9 [g] [REP] [+]` **Ans: 4:9, 8:18, 12:27, 16:36**

#2: In a ratio table where A : B = 4 : 9, find B when A = 28.
SC-6 Keys: `28 [ENTER] 4 [÷] 9 [×]` **Ans: 63**

#3: In a ratio table where X : Y = 5 : 12, find X when Y = 60.
SC-6 Keys: `60 [ENTER] 12 [÷] 5 [×]` **Ans: 25**

#4: A printer prints 15 pages every 2 minutes. How many pages in 8 minutes?
SC-6 Keys: `8 [ENTER] 2 [÷] 15 [×]` **Ans: 60 pages**

#5: Hot cocoa uses 3 tbsp cocoa for every 8 oz milk. How much milk for 18 tbsp cocoa?
SC-6 Keys: `18 [ENTER] 3 [÷] 8 [×]` **Ans: 48 oz milk**

#6: 8 markers cost $12. Scale down to 2 markers, then scale up to find the cost of 10 markers.
SC-6 Keys: `12 [ENTER] 4 [÷] 5 [×]` **Ans: $15**

#7: A heart beats 21 times in 15 seconds. How many beats in 60 seconds (1 minute)?
SC-6 Keys: `60 [ENTER] 15 [÷] 21 [×]` **Ans: 84 beats**

#8: Spot the Glitch: For 3:5, Sam added 4 to both sides and wrote 7:9. Test with [g] [a:b]—are they equal?
SC-6 Keys: `7 [ENTER] 9 [g] [a:b]` **Ans: No! Multiply by 3 -> 9:15**

#9*: 9 smoothies take 12 cups of berries. How many cups of berries for 15 smoothies? Solve 2 ways!
SC-6 Keys: `12 [ENTER] 3 [÷] 5 [×]` **Ans: 20 cups (Way 2: 12÷9 × 15..**

#10*: An e-bike travels 14 miles on 35% battery. How far can it travel on 100% battery?
SC-6 Keys: `14 [ENTER] 7 [÷] 20 [×]` **Ans: 40 miles (Way 2: 2 mi per..**

SEC 3.3: UNIT RATES & BEST-BUY SHOWDOWNS (PLUS [X⇔Y] SWAP RESCUE) [COUNT-01]

Method: Unit Rate Method — Dividing Both Terms by the Second Term to Find Amo..

#1: Find the unit speed in miles per hour: 195 miles in 3 hours.
SC-6 Keys: `195 [ENTER] 3 [÷]` **Ans: 65 mph**

#2: You typed 4 [ENTER] 36 for $36 in 4 hours. Press [x↔y] [÷] to find the hourly pay!
SC-6 Keys: `4 [ENTER] 36 [x⇔y] [÷]` **Ans: $9 / hour**

#3: Find the unit price per pound: $7.80 for 6 pounds of apples.
SC-6 Keys: `7.80 [ENTER] 6 [÷]` **Ans: $1.30 / lb**

#4: A 3D printer uses 90 grams of filament in 6 hours. How many grams per hour?
SC-6 Keys: `90 [ENTER] 6 [÷]` **Ans: 15 g / hr**

#5: Best Buy: Cereal A is $4.50 for 15 oz. Cereal B is $5.04 for 18 oz. Find both unit prices!
SC-6 Keys: `4.50 [ENTER] 15 [÷]; 5.04 [ENTER] 18.` **Ans: Cereal B ($0.28/oz vs $0.3..**

#6: Best Buy: 6-pack of pens for $5.10 vs. 10-pack of pens for $8.20.
SC-6 Keys: `5.10 [ENTER] 6 [÷]; 8.20 [ENTER] 10 [÷]` **Ans: 10-pack ($0.82 vs $0.85)**

#7: A train travels 210 miles in 3.5 hours. At this rate, how far does it travel in 5 hours?
SC-6 Keys: `210 [ENTER] 3.5 [÷] 5 [×]` **Ans: 300 miles**

#8: Pump A fills 45 gallons in 5 min. Pump B fills 64 gallons in 8 min. Which pump is faster?
SC-6 Keys: `45 [ENTER] 5 [÷]; 64 [ENTER] 8 [÷]` **Ans: Pump A (9 gal/min vs 8 gal..**

#9*: An extruder pushes 45 mm of filament in 15 seconds. How much in 60 seconds (1 min)? Solve 2 ways!
SC-6 Keys: `45 [ENTER] 15 [÷] 60 [×]` **Ans: 180 mm (Way 2: 60s is 4×15..**

#10*: Car A drives 168 miles on 6 gallons. Car B drives 232 miles on 8 gallons. Which car gets more mpg?
SC-6 Keys: `168 [ENTER] 6 [÷]; 232 [ENTER] 8 [÷]` **Ans: Car B (29 mpg vs 28 mpg)**

SEC 3.4: PERCENTS, FRACTIONS & DECIMALS — THE 3-COSTUME SWITCHER [FRAC-01]

Method: Graphing Ratio Tables as Ordered Pairs (x, y) & Comparing Steepness (..

#1: Write 3/5 as a decimal and as a percent.
SC-6 Keys: `[.] 3 [.] 5 [ENTER] [FDISP]` **Ans: 0.6 and 60%**

#2: Write 9/25 as a decimal and as a percent.
SC-6 Keys: `[.] 9 [.] 25 [ENTER] [FDISP]` **Ans: 0.36 and 36%**

#3: Write 45% as a decimal and as a simplified fraction ([FDISP]).
SC-6 Keys: `45 [ENTER] 100 [÷] [FDISP]` **Ans: 0.45 and 9/20**

#4: Write 0.125 as a percent and as a simplified fraction ([FDISP]).
SC-6 Keys: `0.125 [ENTER] [FDISP]` **Ans: 12.5% and 1/8**

#5: Find 30% of 80 using the [f] [%] key: 80 [ENTER] 30 [f] [%].
SC-6 Keys: `80 [ENTER] 30 [f] [%]` **Ans: 24**

#6: Find 15% of 60.
SC-6 Keys: `60 [ENTER] 15 [f] [%]` **Ans: 9**

#7: You answered 21 out of 25 questions right. What is your score as a percent?
SC-6 Keys: `21 [ENTER] 25 [÷]` **Ans: 84%**

#8: Convert 150% into a decimal and a mixed fraction.
SC-6 Keys: `150 [ENTER] 100 [÷] [FDISP]` **Ans: 1.5 and 1 1/2**

#9*: Order 5/8, 60%, and 0.61 from least to greatest by converting all 3 to decimals (and percents!).
SC-6 Keys: `[.] 5 [.] 8 [FDISP] -> 0.625` **Ans: 60%, 0.61, 5/8 (Way 2: 60%..**

#10*: A phone battery is at 35% of its 4,000 mAh capacity. How many mAh remain? Solve 2 ways!
SC-6 Keys: `4000 [ENTER] 35 [f] [%]` **Ans: 1,400 mAh (Way 2: 7/20 × 4..**

SEC 3.5: DISCOUNTS, PERCENT CHANGE ([G] [%CHG]) & UNIT CONVERSIONS [PLACE-01]

Method: Unit Analysis (Multiplying by a Conversion Factor That Cancels Units)..

#1: An $80 pair of shoes is 30% off. Find the discount (X) and sale price ([-]).
SC-6 Keys: `80 [ENTER] 30 [f] [%] [-]` **Ans: Discount $24; Sale $56**

#2: A $40 restaurant meal has a 15% tip. Find the tip and total bill ([+]).
SC-6 Keys: `40 [ENTER] 15 [f] [%] [+]` **Ans: Tip $6; Total $46**

#3: Find the percent change from 50 to 65 using [g] [%CHG].
SC-6 Keys: `50 [ENTER] 65 [g] [%CHG]` **Ans: +30%**

#4: Find the percent change from 80 down to 60 using [g] [%CHG].
SC-6 Keys: `80 [ENTER] 60 [g] [%CHG]` **Ans: -25%**

#5: Convert 10 inches to centimeters using [f] [->cm] (2.54 cm = 1 in).
SC-6 Keys: `10 [f] [->cm]` **Ans: 25.4 cm**

#6: Convert 50.8 centimeters back to inches using [g] [->in] ([g][1]).
SC-6 Keys: `50.8 [g] [->in]` **Ans: 20 in**

#7: Convert 11 pounds to kilograms using [f] [->kg] ([f][3]).
SC-6 Keys: `11 [f] [->kg]` **Ans: 4.99 kg**

#8: Reverse Percent: 18 is 25% of what number? (Divide 18 by 0.25!)
SC-6 Keys: `18 [ENTER] 0.25 [÷]` **Ans: 72**

#9*: A $120 skateboard is 20% off. Find the sale price both by subtracting [f][%] AND by multiplying by 80%!
SC-6 Keys: `120 [ENTER] 20 [f] [%] [-]` **Ans: $96 (Way 2: 120 × 0.80 = 96)**

#10*: A 32-inch deck weighs 5 lbs. Convert length to cm ([f][->cm]) and weight to kg ([f][->kg]). Verify with ×2.54!
SC-6 Keys: `32 [f] [->cm]; 5 [f] [->kg]` **Ans: 81.28 cm, 2.27 kg (Way 2:..**

Unit 4: Algebraic Expressions & Properties — Complete Key

Big Ideas Math Grade 6 — Chapter 3 / Chapter 5 (Algebraic Expressions) • Manipulative Kits: Algebra Boxes & Factoring Stage (ALGB-01) + Counting Tiles (COUNT-01)

Target Misconception: Students panic when letters appear, erroneously combine 3x + 2 into 5x or x² + x into x³, and don't see why 3(x + 4) = 3x + 12.

Hands-On Block Remediation: Show that a variable x is literally a physical green x-rod (42×14mm) whereas 1 is a small yellow unit cube (14×14mm) and x² is a large blue flat (42×42mm)—different physical shapes cannot be counted as one pile! Build rectangles on the L-frame stage for the Distributive Property.



Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!

SEC 4.1: EVALUATING ALGEBRAIC EXPRESSIONS — VARIABLE BOXES ON THE STACK [ALGB-01]

Method: 10×10 Hundredths Grid ('Per 100'), Decimal-Point Shift & Fraction Ben..

#1: Evaluate 6x - 9 when x = 7.
SC-6 Keys: 6 [ENTER] 7 [×] 9 [-] **Ans: 33**

#2: Evaluate 5n + 14 when n = 8.
SC-6 Keys: 5 [ENTER] 8 [×] 14 [+] **Ans: 54**

#3: Evaluate 40 - 3m when m = 9. (Park 40 on Y first!)
SC-6 Keys: 40 [ENTER] 3 [ENTER] 9 [×] [-] **Ans: 13**

#4: Evaluate x² + 11 when x = 6.
SC-6 Keys: 6 [x²] 11 [+] **Ans: 47**

#5: Evaluate 2y² - 5 when y = 4. (Square 4 before multiplying by 2!)
SC-6 Keys: 4 [x²] 2 [×] 5 [-] **Ans: 27**

#6: Evaluate 7a + 3b when a = 4 and b = 9.
SC-6 Keys: 7 [ENTER] 4 [×] 3 [ENTER] 9 [×] [+] **Ans: 55**

#7: Evaluate (x + y) ÷ 4 when x = 19 and y = 13.
SC-6 Keys: 19 [ENTER] 13 [+] 4 [÷] **Ans: 8**

#8: Evaluate 5k - 2.4 when k = 3.5.
SC-6 Keys: 5 [ENTER] 3.5 [×] 2.4 [-] **Ans: 15.1**

#9*: Skate park rental costs $8 flat plus $4.50 per hour (8 + 4.50h). Find the cost for 4 hours 2 ways!
SC-6 Keys: 4.50 [ENTER] 4 [×] 8 [+] **Ans: $26 (Way 2: 8+4.5+4.5+4.5+..**

#10*: Surface area of a cube is 6s². Find the surface area when edge s = 5 cm.
SC-6 Keys: 5 [x²] 6 [×] **Ans: 150 cm² (Way 2: 6 faces ×..**

SEC 4.2: COMBINING LIKE TERMS — WHY X² FLATS, X RODS & 1 CUBES NEVER MIX [ALGB-01]

Method: Ordering Fractions, Decimals & Percents by Converting to a Common For..

#1: Sort and simplify: 7x + 4 + 2x + 9.
SC-6 Keys: 7 [ENTER] 2 [+]; 4 [ENTER] 9 [+] **Ans: 9x + 13**

#2: Simplify: 8x + 11 - 3x - 4.
SC-6 Keys: 8 [ENTER] 3 [-]; 11 [ENTER] 4 [-] **Ans: 5x + 7**

#3: Simplify: 9y - y + 6. (Remember: -y means -1y!)
SC-6 Keys: 9 [ENTER] 1 [-] **Ans: 8y + 6**

#4: Simplify: 12a + 5 - 4a + 8.
SC-6 Keys: 12 [ENTER] 4 [-]; 5 [ENTER] 8 [+] **Ans: 8a + 13**

#5: Sort all 3 tile shapes: 4x² + 3x + 2x² + 5x + 7.
SC-6 Keys: 4+2=6; 3+5=8 **Ans: 6x² + 8x + 7**

#6: Simplify: 6n² + 9n - 2n² - 4n + 10.
SC-6 Keys: 6-2=4; 9-4=5 **Ans: 4n² + 5n + 10**

#7: Simplify with decimals: 3.5x + 4.2 + 1.8x - 1.5.
SC-6 Keys: 3.5 [ENTER] 1.8 [+]; 4.2 [ENTER] 1.5.. **Ans: 5.3x + 2.7**

#8: Spot the Glitch: Leo simplified 5x + 3 to 8x. Test x = 10 on your SC-6 to prove 5x + 3 != 8x!
SC-6 Keys: 5(10)+3=53 vs 8(10)=80 **Ans: 53 != 80; 5x + 3 is alread..**

#9*: A rectangle has two sides of (2x + 3) and two sides of (x + 5). Write its simplified perimeter & test x=10!
SC-6 Keys: 2+2+1+1=6x; 3+3+5+5=16 **Ans: 6x + 16 (At x=10: both giv..**

#10*: A triangle has sides (3x + 2), (2x + 7), and (4x - 1). Write the simplified perimeter & test x=5!
SC-6 Keys: 3+2+4=9x; 2+7-1=8 **Ans: 9x + 8 (At x=5: 17+17+19 =..**

SEC 4.3: THE DISTRIBUTIVE PROPERTY — ONE SPLIT RECTANGLE, TWO WAYS [ALGB-01]

Method: Double Number Line, 10% Mental Benchmark & Percent Equation (Part = P..

#1: Compute 7 × (20 + 3) both ways on your SC-6. Write the two partial products.
SC-6 Keys: 7 [ENTER] 20 [×] 7 [ENTER] 3 [×] [+] **Ans: 140 + 21 = 161**

#2: Build 4 rows of (x + 5) and write the distributed expression for 4(x + 5).
SC-6 Keys: 4 [ENTER] 5 [×] **Ans: 4x + 20**

#3: Use the Distributive Property to expand 6(3x + 4).
SC-6 Keys: 6×3=18; 6×4=24 **Ans: 18x + 24**

#4: Expand 8(2y - 5). Watch the subtraction sign!
SC-6 Keys: 8×2=16; 8×5=40 **Ans: 16y - 40**

#5: Mental Math Split: Compute 9 × 43 by splitting 43 into (40 + 3).
SC-6 Keys: 9 [ENTER] 40 [×] 9 [ENTER] 3 [×] [+] **Ans: 387**

#6: Distribute and combine like terms: 3(4x + 2) + 5x.
SC-6 Keys: 3×4+5=17; 3×2=6 **Ans: 17x + 6**

#7: Distribute and combine: 4(3n + 5) - 7.
SC-6 Keys: 4×3=12; 20-7=13 **Ans: 12n + 13**

#8: Spot the Glitch: Mia wrote 5(x + 6) = 5x + 6. What did she forget to multiply by 5?
SC-6 Keys: 5 [ENTER] 6 [×] **Ans: 5x + 30 (Must multiply 6 b..**

#9*: Simplify 5(2x + 3) + 4x AND prove your answer matches the original when x = 2!
SC-6 Keys: 5×2+4=14; 5×3=15 **Ans: 14x + 15 (At x=2: both equ..**

#10*: 6 friends each buy a burger ($x) and $3.50 fries: 6(x + 3.50). Expand it and evaluate for x = $6.50!
SC-6 Keys: 6 [ENTER] 3.50 [×] -> 6x+21 **Ans: 6x + 21; $60 total**

SEC 4.4: FACTORING EXPRESSIONS WITH GCF — REVERSE THE RECTANGLE [ALGB-01]

Method: Percent Proportion (Part / Whole = p / 100) & Tape Diagram to Find th..

#1: Factor 12 + 20 into the form a(b + c) using [GCF].
SC-6 Keys: 12 [ENTER] 20 [GCF] -> 4 **Ans: 4(3 + 5)**

#2: Factor 8x + 12 using [GCF].
SC-6 Keys: 8 [ENTER] 12 [GCF] -> 4 **Ans: 4(2x + 3)**

#3: Factor 15y - 25 using [GCF].
SC-6 Keys: 15 [ENTER] 25 [GCF] -> 5 **Ans: 5(3y - 5)**

#4: Factor 24x + 36 completely using [GCF].
SC-6 Keys: 24 [ENTER] 36 [GCF] -> 12 **Ans: 12(2x + 3)**

#5: Factor 32n - 48 completely using [GCF].
SC-6 Keys: 32 [ENTER] 48 [GCF] -> 16 **Ans: 16(2n - 3)**

#6: Factor 42a + 14 completely. (Careful: what is 14 ÷ 14?)
SC-6 Keys: 42 [ENTER] 14 [GCF] -> 14 **Ans: 14(3a + 1)**

#7: Factor 36m - 60 completely using [GCF].
SC-6 Keys: 36 [ENTER] 60 [GCF] -> 12 **Ans: 12(3m - 5)**

#8: Spot the Glitch: Sam factored 24x + 36 as 6(4x + 6). Why isn't it factored completely?
SC-6 Keys: 4 [ENTER] 6 [GCF] -> 2 left! **Ans: 12(2x + 3) (4 and 6 still..**

#9*: A rectangular banner has area (28x + 42) sq in and height 14 in. Factor with [GCF] to find its width!
SC-6 Keys: 28 [ENTER] 42 [GCF] -> 14 **Ans: Width = 2x + 3 (Way 2: 14(..**

#10*: A square garden has perimeter (36x + 48) ft. Factor out 4 to find the length of 1 side!
SC-6 Keys: 36 [ENTER] 4 [÷]; 48 [ENTER] 4 [÷] **Ans: Side = 9x + 12 (4(9x + 12))**

Unit 5: Areas of Polygons & Elapsed Time — Complete Key

Big Ideas Math Grade 6 — Chapter 4 (Areas of Polygons) • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Planetary Geared Clock (TIME-01)

Target Misconception: Students multiply the slanted side of a parallelogram instead of the vertical height, forget to divide by 2 for triangles, lose track of multi-piece composite areas, and subtract clock times in base-10 instead of base-60.

Hands-On Block Remediation: Slide the corner triangle of a parallelogram to form a straight [f] [RECT] rectangle, use the dedicated [g] [TRI] key ([g][7]) which divides by 2 automatically, accumulate composite chunks with [Σ+], and use the Planetary Clock (TIME-01) + [CLK] key for base-60 time.

Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!



SEC 5.1: AREAS OF PARALLELOGRAMS & RECTANGLES ([F] [RECT]) [COUNT-01]

Method: Substituting Values for Variables in Expressions & Applying Order of..

#1: Find the area of a parallelogram with base b = 15 m, vertical height h = 9 m, and slanted side 11 m.
SC-6 Keys: 15 [ENTER] 9 [f][RECT] **Ans: 135 m² (Ignore 11m!)**

#2: Find the area of a rectangle measuring 18 ft by 12 ft using [f] [RECT].
SC-6 Keys: 18 [ENTER] 12 [f][RECT] **Ans: 216 ft²**

#3: Find the area of a parallelogram with b = 8.5 m and vertical height h = 6.4 m.
SC-6 Keys: 8.5 [ENTER] 6.4 [f][RECT] **Ans: 54.4 m²**

#4: Find the area of a square tile with side 13 cm.
SC-6 Keys: 13 [x²] **Ans: 169 cm²**

#5: A parallelogram has base b = 6 1/2 in and height h = 4 in. Find its area.
SC-6 Keys: 6 [.] 1 [.] 2 [ENTER] 4 [f][RECT] **Ans: 26 in²**

#6: Missing Height: A parallelogram has Area = 108 cm² and base b = 12 cm. Find h.
SC-6 Keys: 108 [ENTER] 12 [÷] **Ans: 9 cm**

#7: Missing Base: A parallelogram has Area = 135 ft² and vertical height h = 7.5 ft. Find b.
SC-6 Keys: 135 [ENTER] 7.5 [÷] **Ans: 18 ft**

#8: Spot the Glitch: A parallelogram has b = 10, vertical h = 6, slanted side = 7.5. Sam got 75. Fix it!
SC-6 Keys: 10 [ENTER] 6 [f][RECT] **Ans: 60 (Sam multiplied slanted..**

#9*: A slanted parking space has base 9 ft and straight length 18.5 ft. How much asphalt covers 4 spaces?
SC-6 Keys: 9 [ENTER] 18.5 [f][RECT] 4 [×] **Ans: 666 ft² (Way 2: base 36 ×..**

#10*: A rectangular patio (15 ft × 12 ft) has a parallelogram planter (b=4 ft, h=3 ft) inside. Find the remaining patio..
SC-6 Keys: 15 [ENTER] 12 [f][RECT] 4 [ENTER] 3 [. **Ans: 168 ft² (Way 2: 180 - 12 =..**

SEC 5.2: AREAS OF TRIANGLES — HALF A RECTANGLE ([G] [TRI]) [COUNT-01]

Method: Translating Verbal Phrases into Algebraic Expressions & Identifying T..

#1: Find the area of a triangle with base b = 14 ft and height h = 18 ft using [g] [TRI].
SC-6 Keys: 14 [ENTER] 18 [g][TRI] **Ans: 126 ft²**

#2: Find the area of a right triangle with base 9 ft, vertical height 4 ft, and slanted hypotenuse 9.85 ft.
SC-6 Keys: 9 [ENTER] 4 [g][TRI] **Ans: 18 ft² (Ignore 9.85!)**

#3: Find the area of a triangle with b = 16 cm and h = 11 cm.
SC-6 Keys: 16 [ENTER] 11 [g][TRI] **Ans: 88 cm²**

#4: Find the area of a triangle with decimal dimensions: b = 12.5 m, h = 8.4 m.
SC-6 Keys: 12.5 [ENTER] 8.4 [g][TRI] **Ans: 52.5 m²**

#5: Find the area of a triangle with fractional dimensions: b = 4 1/2 in, h = 6 in.
SC-6 Keys: 4 [.] 1 [.] 2 [ENTER] 6 [g][TRI] **Ans: 13 1/2 in² (13.5)**

#6: Missing Height: A triangle has Area = 54 cm² and base b = 12 cm. (Double 54 first, then ÷ 12!)
SC-6 Keys: 54 [ENTER] 2 [×] 12 [÷] **Ans: 9 cm**

#7: Missing Base: A triangle has Area = 84 m² and height h = 14 m. Find base b.
SC-6 Keys: 84 [ENTER] 2 [×] 14 [÷] **Ans: 12 m**

#8: Spot the Glitch: Leo found the area of a b=8, h=5 triangle as 40. What did he calculate instead?
SC-6 Keys: 8 [ENTER] 5 [g][TRI] **Ans: 20 (40 is the full rectang..**

#9*: A sailboat has a main sail (b=12 ft, h=25 ft) and jib sail (b=8 ft, h=15 ft). Find total sail area 2 ways!
SC-6 Keys: 12 [ENTER] 25 [g][TRI] 8 [ENTER] 15 [. **Ans: 210 ft² (Way 2: (300+120)÷..**

#10*: A triangular gable window has Area = 36 ft² and base 9 ft. Find its height and check with [g] [TRI]!
SC-6 Keys: 36 [ENTER] 2 [×] 9 [÷] **Ans: 8 ft (Way 2: 9 [ENTER] 8 [..**

SEC 5.3: TRAPEZOIDS & COMPOSITE FIGURES WITH THE [Σ+] RUNNING LEDGER [COUNT-01]

Method: Commutative (Reorder) & Associative (Regroup) Properties of Addition..

#1: Find the area of a trapezoid with b1 = 10 cm, b2 = 14 cm, and h = 6 cm using (b1+b2) [ENTER] h [g] [TRI].
SC-6 Keys: 10 [ENTER] 14 [+] 6 [g][TRI] **Ans: 72 cm²**

#2: Find the area of a trapezoid with b1 = 18 ft, b2 = 12 ft, and h = 9 ft.
SC-6 Keys: 18 [ENTER] 12 [+] 9 [g][TRI] **Ans: 135 ft²**

#3: L-Shape Room: Split into Room A (10 × 6) and Room B (8 × 4). Sum with [Σ+] and [f] [Σx].
SC-6 Keys: 10 [ENTER] 6 [f][RECT] [Σ+] 8 [ENTER] 4 [f][RECT] [Σ+].. **Ans: 92**

#4: Banner Arrow: A 15 × 6 rectangle attached to a triangle (b = 10, h = 6). Find total area.
SC-6 Keys: 15 [ENTER] 6 [f][RECT] [Σ+] 10 [ENTER] 6 [g][TRI] [Σ+] [.. **Ans: 120**

#5: Find the area of a trapezoid with decimal bases b1 = 8.5 m, b2 = 5.5 m, and h = 4.2 m.
SC-6 Keys: 8.5 [ENTER] 5.5 [+] 4.2 [g][TRI] **Ans: 29.4 m²**

#6: Cutout Frame: A 10 in × 8 in frame has a 6 in × 4 in hole cut out. Use [Σ+] and [f] [Σ-] to find the border area!
SC-6 Keys: 10 [ENTER] 8 [f][RECT] [Σ+] 6 [ENTER] 4 [f][RECT] [f][Σ-.. **Ans: 56 in²**

#7: Missing Trapezoid Height: A trapezoid has Area = 90 cm² and bases b1 = 11 cm, b2 = 7 cm. Find h.
SC-6 Keys: 90 [ENTER] 2 [×] 11 [ENTER] 7 [+] [÷] **Ans: 10 cm**

#8: A mini-golf hole is a 12 × 5 ft rectangle plus a trapezoid (b1=5, b2=9, h=4). Find total area.
SC-6 Keys: 60 [Σ+] 5 [ENTER] 9 [+] 4 [g][TRI] [Σ+] [f][Σx] **Ans: 88 ft²**

#9*: Find the area of a trapezoid (b1=10, b2=16, h=8) both with the trapezoid formula AND by splitting into a 10×..
SC-6 Keys: 10 [ENTER] 16 [+] 8 [g][TRI] **Ans: 104 (Way 2: 80 + 24 = 104)**

#10*: A 20 ft × 15 ft lawn has a triangular flowerbed (b=8 ft, h=6 ft) and square shed (5 ft × 5 ft). Find grass area!
SC-6 Keys: 300 [Σ+] 24 [f][Σ-] 25 [f][Σ-] [f][Σx] **Ans: 251 ft²**

SEC 5.4: COORDINATE POLYGONS & PLANETARY CLOCK BASE-60 TIME ([CLK]) [TIME-01]

Method: Distributive Property Area Model — Expanding a(b + c) = ab + ac & Fac..

#1: Find the distance between (3, 4) and (11, 4) on a coordinate grid.
SC-6 Keys: 11 [ENTER] 3 [-] **Ans: 8 units**

#2: A rectangle has corners (1, 2), (10, 2), (10, 8), (1, 8). Find its area.
SC-6 Keys: 10 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][.. **Ans: 54 sq units**

#3: In [CLK] mode, add 1 hr 45 min + 1 hr 35 min (1.45 [ENTER] 1.35 [+]).
SC-6 Keys: [CLK] 1.45 [ENTER] 1.35 [+] **Ans: 3 hr 20 min (3.20)**

#4: Convert 2 hr 30 min (2.30) to decimal hours using [f] [->HR].
SC-6 Keys: 2.30 [f][->HR] **Ans: 2.5 hours**

#5: Convert 4 hr 45 min (4.45) to decimal hours using [f] [->HR].
SC-6 Keys: 4.45 [f][->HR] **Ans: 4.75 hours**

#6: Convert 1.75 decimal hours back into Hours & Minutes using [g] [->HMS].
SC-6 Keys: 1.75 [g][->HMS] **Ans: 1 hr 45 min (1.45)**

#7: In [CLK] mode, find elapsed time from 8:50 AM to 11:25 AM (11.25 [ENTER] 8.50 [-]).
SC-6 Keys: [CLK] 11.25 [ENTER] 8.50 [-] **Ans: 2 hr 35 min (2.35)**

#8: A right triangle on a grid has vertices (2, 1), (10, 1), and (2, 7). Find its area!
SC-6 Keys: 8 [ENTER] 6 [g][TRI] **Ans: 24 sq units**

#9*: You tutor for 3 hr 15 min (3.15) at $16/hr. Convert to decimal hours with [f] [->HR] and find your pay 2 ways!
SC-6 Keys: 3.15 [f][->HR] 16 [×] **Ans: $52.00 (Way 2: 3×16 + ¼×16..**

#10*: Coordinate Park: Vertices (2, 2), (14, 2), (14, 11), (2, 11) where 1 unit = 10 yd. Find perimeter and area!
SC-6 Keys: 12 [ENTER] 9 [f][RECT] **Ans: P = 42 units (420 yd); A =..**

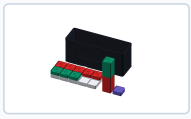
Unit 6: Integers, Rational Numbers & The Coordinate Plane — Complete Key

Big Ideas Math Grade 6 — Chapter 6 • Manipulative Kits: Ten-Frame Bicolor Counters (BASE-10) + Number Rack (RACK-01)

Target Misconception: Students confuse the subtraction operator [-] with the negative sign [+/-], think -18 > -5 because 18 > 5, confuse Opposite ([+/-]) with Absolute Value ([g] [|x|]), and reverse (x, y) coordinates.

Hands-On Block Remediation: Use Yellow (+1) and Red (-1) Zero-Pair counters, test inequalities with [f] [>] and [g] [<], contrast [+/-] (flip sign) with [g] [|x|] (always positive distance from 0), and use the Elevator Rule (walk across x first, ride up/down y second).

Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!



SEC 6.1: POSITIVE & NEGATIVE INTEGERS — ZERO PAIRS & THE [+/-] SIGN FLIPPER [BASE-10]

Method: Positive & Negative Integers on a Number Line & Finding Opposites Acr..

#1: Write the integer for '25 feet below sea level' and find its opposite using [+/-].
SC-6 Keys: `25 [+/-] -> -25; [+/-] -> 25` **Ans: -25; Opposite = +25**

#2: Find the opposite of -42, and then the opposite of the opposite (-[-(-42)]).
SC-6 Keys: `42 [+/-] [+/-] [+/-]` **Ans: +42; and -42**

#3: Zero-Pair Cancellation: A submarine is at -18 m and rises +18 m. Where is it now?
SC-6 Keys: `18 [+/-] [ENTER] 18 [+]` **Ans: 0 m (Sea level)**

#4: Combine 6 Yellow (+1) counters and 6 Red (-1) counters. What is the value?
SC-6 Keys: `6 [ENTER] 6 [+/-] [+]` **Ans: 0**

#5: A hiker starts at -120 ft in a canyon and climbs +120 ft. What is her elevation?
SC-6 Keys: `120 [+/-] [ENTER] 120 [+]` **Ans: 0 ft**

#6: What integer is its own opposite? Test 0 [+/-] on your SC-6!
SC-6 Keys: `0 [+/-]` **Ans: 0**

#7: Temperature starts at 0°F, drops 12°F (12 [+/-]), and rises 19°F. Find the temperature.
SC-6 Keys: `12 [+/-] [ENTER] 19 [+]` **Ans: +7°F**

#8: Golf Par: -3 means 3 strokes under par. If you shoot -3 on Saturday and +3 on Sunday, what is your total?
SC-6 Keys: `3 [+/-] [ENTER] 3 [+]` **Ans: 0 (Even par)**

#9*: Elevator starts at Floor 0, goes up 14, down 20 (20 [+/-]), and up 9. Find the floor 2 ways!
SC-6 Keys: `14 [ENTER] 20 [+/-] [+] 9 [+]` **Ans: Floor 3 (Way 2: (14+9) - 2..**

#10*: Game XP: You gain 40 XP, lose 65 XP, and gain 25 XP. Did you break even? Prove 2 ways!
SC-6 Keys: `40 [ENTER] 65 [+/-] [+] 25 [+]` **Ans: 0 XP (Break even! 40+25 =..**

SEC 6.2: COMPARING & ORDERING INTEGERS AND RATIONAL NUMBERS [RACK-01]

Method: Comparing & Ordering Integers and Rational Numbers on a Horizontal/Ve..

#1: Which temperature is warmer (greater): -12°C or -5°C? Test with [g] [<].
SC-6 Keys: `12 [+/-] [ENTER] 5 [+/-] [g] [<]` **Ans: -5°C (-12 < -5)**

#2: Compare -9 and -2 using > or <.
SC-6 Keys: `9 [+/-] [ENTER] 2 [+/-] [g] [<]` **Ans: -9 < -2**

#3: Compare -15 and 0 using > or <.
SC-6 Keys: `15 [+/-] [ENTER] 0 [g] [<]` **Ans: -15 < 0**

#4: Order the integers from least to greatest: 4, -7, 0, -2, -10.
SC-6 Keys: `Left-to-right on number line` **Ans: -10, -7, -2, 0, 4**

#5: Compare -3/4 and -0.6. (Hint: -3/4 = -0.75. Which is further right?)
SC-6 Keys: `0.75 [+/-] [ENTER] 0.6 [+/-] [g] [<]` **Ans: -3/4 < -0.6 (-0.6 is great..**

#6: Compare -2 1/2 and -2.3 using > or <.
SC-6 Keys: `2.5 [+/-] [ENTER] 2.3 [+/-] [g] [<]` **Ans: -2 1/2 < -2.3**

#7: Compare -5/8 and -7/8.
SC-6 Keys: `[.] 5 [.] 8 [+/-] [ENTER] [.] 7 [.] 8..` **Ans: -5/8 > -7/8**

#8: Spot the Glitch: Leo says -4.5 > -4.1 because 4.5 > 4.1. Explain why -4.1 is actually greater!
SC-6 Keys: `4.5 [+/-] [ENTER] 4.1 [+/-] [g] [<]` **Ans: -4.1 is closer to 0 (furth..**

#9*: Order -1.5, 1 1/4, -1 3/4, and -0.5 from least to greatest. Verify both as decimals AND with [g] [<]!
SC-6 Keys: `-1.75, -1.5, -0.5, 1.25` **Ans: -1 3/4, -1.5, -0.5, 1 1/4**

#10*: Three divers are at -14.8 m, -14 1/2 m, and -15.2 m. Who is deepest (least) and who is shallowest (greatest..
SC-6 Keys: `-15.2 < -14.8 < -14.5` **Ans: Deepest: -15.2 m; Shallowe..**

SEC 6.3: ABSOLUTE VALUE — DISTANCE FROM ZERO IS ALWAYS POSITIVE ([G] [|X|]) [RACK-01]

Method: Absolute Value |x| as Distance from Zero (Always Non-Negative) & Eval..

#1: Find |-27| using [g] [|x|].
SC-6 Keys: `27 [+/-] [g] [|x|]` **Ans: 27**

#2: Evaluate |-27| + |13|.
SC-6 Keys: `27 [+/-] [g] [|x|] [ENTER] 13 [+]` **Ans: 40**

#3: Evaluate |-4.8| + |-3.2|.
SC-6 Keys: `4.8 [+/-] [g] [|x|] [ENTER] 3.2 [+/-]..` **Ans: 8.0**

#4: Evaluate |-35| - |-19|.
SC-6 Keys: `35 [+/-] [g] [|x|] [ENTER] 19 [+/-] [g..` **Ans: 16**

#5: Compare |-12| and |9| using > or <. (Which is further from 0?)
SC-6 Keys: `12 > 9` **Ans: |-12| > |9| (12 > 9)**

#6: Evaluate -|-24|. (Find |-24| first, then press [+/-] for the outside minus!)
SC-6 Keys: `24 [+/-] [g] [|x|] [+/-]` **Ans: -24**

#7: Evaluate 50 - |-18|. Find |-18| first, then subtract from 50.
SC-6 Keys: `50 [ENTER] 18 [+/-] [g] [|x|] [-]` **Ans: 32**

#8: Order from least to greatest: |-8|, -6, |-3|, 0, -|-9|.
SC-6 Keys: `Values: 8, -6, 3, 0, -9` **Ans: -|-9|, -6, 0, |-3|, |-8|**

#9*: A diver is at -32 ft and a seagull is at +25 ft. Which is closer to sea level (0), and by how many feet?
SC-6 Keys: `32 [+/-] [g] [|x|] [ENTER] 25 [-]` **Ans: Seagull is closer by 7 ft..**

#10*: Two accounts have balances of -$45 and -$28. Which account has the greater debt (|-45| vs |-28|), and whi..
SC-6 Keys: `|-45|=45 > |-28|=28` **Ans: Greater debt: -$45 ($45);..**

SEC 6.4: THE 4-QUADRANT COORDINATE PLANE & AXIS REFLECTIONS [COUNT-01]

Method: Plotting (x, y) in All Four Quadrants (I, II, III, IV) & Reflecting A..

#1: Name the Quadrant (I, II, III, or IV) for point (7, -4).
SC-6 Keys: `(+, -) -> Right, Down` **Ans: Quadrant IV**

#2: Name the Quadrant for point (-9, -2).
SC-6 Keys: `(-, -) -> Left, Down` **Ans: Quadrant III**

#3: Name the Quadrant for point (-5, 11).
SC-6 Keys: `(-, +) -> Left, Up` **Ans: Quadrant II**

#4: Reflect (6, 9) across the y-axis. (Flip the x-coordinate with [+/-]!)
SC-6 Keys: `6 [+/-] -> -6` **Ans: (-6, 9)**

#5: Reflect (6, 9) across the x-axis. (Flip the y-coordinate with [+/-]!)
SC-6 Keys: `9 [+/-] -> -9` **Ans: (6, -9)**

#6: Reflect (-3, -7) across the x-axis.
SC-6 Keys: `7 [+/-] [+/-] -> +7` **Ans: (-3, 7)**

#7: Reflect (-8, 4) across the y-axis, and then across the x-axis.
SC-6 Keys: `Flip x -> (8,4); flip y -> (8,-4)` **Ans: (8, -4)**

#8: On the Fence: Which axis does (0, -9) sit on, and which axis does (-7, 0) sit on?
SC-6 Keys: `0 sideways vs 0 up/down` **Ans: (0,-9) on y-axis; (-7,0) o..**

#9*: Start at (4, -5). Walk 7 units left and 8 units up. Where do you land, and in which quadrant?
SC-6 Keys: `4 [ENTER] 7 [-]; 5 [+/-] [ENTER] 8 [+]` **Ans: (-3, 3) in Quadrant II**

#10*: Points A(-4, 2), B(4, 2), and C(4, -5) are 3 corners of a rectangle. Find the 4th corner D!
SC-6 Keys: `Match x of A (-4) & y of C (-5)` **Ans: D(-4, -5) in Quadrant III**

SEC 6.5: DISTANCE BETWEEN POINTS ON THE COORDINATE PLANE ([G] [|X|]) [COUNT-01]

Method: Finding Horizontal & Vertical Distances on the Coordinate Plane Using..

#1: Find the horizontal distance between (-8, 5) and (11, 5).
SC-6 Keys: `8 [+/-] [ENTER] 11 [-] [g] [|x|]` **Ans: 19 units**

#2: Find the vertical distance between (-3, -4) and (-3, -15).
SC-6 Keys: `4 [+/-] [ENTER] 15 [+/-] [-] [g] [|x|]` **Ans: 11 units**

#3: Find the vertical distance between (7, 9) and (7, -6).
SC-6 Keys: `9 [ENTER] 6 [+/-] [-] [g] [|x|]` **Ans: 15 units**

#4: Find the horizontal distance between (-14, 2) and (-5, 2).
SC-6 Keys: `14 [+/-] [ENTER] 5 [+/-] [-] [g] [|x|]` **Ans: 9 units**

#5: Find the distance between (-3.5, 4) and (6.5, 4).
SC-6 Keys: `3.5 [+/-] [ENTER] 6.5 [-] [g] [|x|]` **Ans: 10 units**

#6: Find the distance between (2, -4 1/2) and (2, 3 1/2).
SC-6 Keys: `4.5 [+/-] [ENTER] 3.5 [-] [g] [|x|]` **Ans: 8 units**

#7: A line segment starts at (-6, 3) and goes 10 units right. What are the coordinates of the right endpoint?
SC-6 Keys: `6 [+/-] [ENTER] 10 [+]` **Ans: (4, 3)**

#8: Spot the Glitch: For (-5, 2) to (8, 2), Sam subtracted 8 - 5 = 3. Why is the distance 13, not 3?
SC-6 Keys: `5 [+/-] [ENTER] 8 [-] [g] [|x|]` **Ans: 13 units (They are on oppo..**

#9*: A treasure map rectangle has corners (-4, 6), (5, 6), (5, -2), (-4, -2). Find its perimeter & area 2 ways!
SC-6 Keys: `Width 9, Height 8` **Ans: Perimeter = 34, Area = 72**

#10*: On a city grid (1 unit = 0.5 mile), you walk from (-6, -2) to (-6, 8) to (4, 8). How many miles did you walk?
SC-6 Keys: `10 + 10 = 20 units × 0.5` **Ans: 20 units = 10 miles**

Unit 7: Equations & Inequalities — Complete Key

Big Ideas Math Grade 6 — Chapter 7 • Manipulative Kits: Algebra Boxes & Stage (ALGB-01) + Number Rack (RACK-01)

Target Misconception: Students treat '=' as 'write the answer', translate '8 less than n' backwards as 8 - n, resent writing -14 under both sides of x + 14 = 39, and confuse open vs. solid inequality circles.

Hands-On Block Remediation: Model '=' as a Balance Scale using Algebra Boxes & Stage (ALGB-01). Solve 1-step equations with inverse operations on both sides and check the answer with [f] [LASTX], and test inequalities with [f] [≥] and [g] [≤].



Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!

SEC 7.1: WRITING EQUATIONS & TESTING SOLUTIONS — THE BALANCE SCALE RULE [ALGB-01]

Method: Writing Equations in One Variable & Testing Solutions by Substitution..

#1: Test whether x = 6 or x = 8 solves 5x - 7 = 33.
SC-6 Keys: `5 [ENTER] 8 [×] 7 [-]` **Ans: x = 8 (5(8)-7 = 33)**

#2: Test whether y = 12 or y = 15 solves y / 3 + 4 = 9.
SC-6 Keys: `15 [ENTER] 3 [÷] 4 [+]` **Ans: y = 15 (15/3 + 4 = 9)**

#3: Translate into an equation: 'The sum of a number x and 14 is 52.'
SC-6 Keys: `x + 14 = 52` **Ans: x + 14 = 52 (x = 38)**

#4: Translate into an equation: '6 times a number k equals 84.'
SC-6 Keys: `6k = 84` **Ans: 6k = 84 (k = 14)**

#5: Translate: '9 less than a number w is 17.' (Watch the order!)
SC-6 Keys: `w - 9 = 17` **Ans: w - 9 = 17 (w = 26)**

#6: Translate: 'The quotient of a number m and 5 is 13.'
SC-6 Keys: `m / 5 = 13` **Ans: m/5 = 13 (m = 65)**

#7: 4 pizza pies (p) plus a $6 delivery fee total $54. Write the equation and test p = 12!
SC-6 Keys: `4 [ENTER] 12 [×] 6 [+]` **Ans: 4p + 6 = 54; p = $12**

#8: Spot the Glitch: For '12 less than x is 20', Sam wrote 12 - x = 20. Fix the equation and solve!
SC-6 Keys: `x - 12 = 20` **Ans: x - 12 = 20 (x = 32)**

#9*: A streaming pass costs $10 plus $3 per movie (10 + 3m = 28). Find m by testing AND by undoing!
SC-6 Keys: `28 [ENTER] 10 [-] 3 [÷]` **Ans: m = 6 movies (10 + 3(6) =..**

#10*: Three friends split a restaurant bill b equally and each pays $14.50. Write the equation and find b!
SC-6 Keys: `14.50 [ENTER] 3 [×]` **Ans: b/3 = 14.50; b = $43.50**

SEC 7.2: SOLVING ONE-STEP ADDITION & SUBTRACTION EQUATIONS ([F] [LASTX]) [ALGB-01]

Method: Subtraction & Addition Properties of Equality (Inverse Operations on..

#1: Solve x + 28 = 73 and prove with [f] [LASTX] [+].
SC-6 Keys: `73 [ENTER] 28 [-]` **Ans: x = 45**

#2: Solve y - 34 = 58 and prove with [f] [LASTX] [-].
SC-6 Keys: `58 [ENTER] 34 [+]` **Ans: y = 92**

#3: Solve 19 + n = 64.
SC-6 Keys: `64 [ENTER] 19 [-]` **Ans: n = 45**

#4: Solve k - 47 = 29.
SC-6 Keys: `29 [ENTER] 47 [+]` **Ans: k = 76**

#5: Solve w - 19.6 = 34.8.
SC-6 Keys: `34.8 [ENTER] 19.6 [+]` **Ans: w = 54.4**

#6: Solve p + 8.75 = 25.4.
SC-6 Keys: `25.4 [ENTER] 8.75 [-]` **Ans: p = 16.65**

#7: Solve m + 3/8 = 5/6 on your SC-6 and check with [f] [LASTX] [+].
SC-6 Keys: `[.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]` **Ans: m = 11/24**

#8: Solve a - 1 1/4 = 3 1/2.
SC-6 Keys: `3 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [+]` **Ans: a = 4 3/4**

#9*: After spending $18.50 on a controller skin, you have $26.75 left (m - 18.50 = 26.75). Find starting money m!
SC-6 Keys: `26.75 [ENTER] 18.50 [+]` **Ans: $45.25 (Check: 45.25 - 18...**

#10*: A drone climbs 42.5 ft to reach 110 ft (h + 42.5 = 110). Find its starting height h and prove with [f][LASTX]!
SC-6 Keys: `110 [ENTER] 42.5 [-]` **Ans: 67.5 ft (Check: 67.5 + 42...**

SEC 7.3: SOLVING ONE-STEP MULTIPLICATION & DIVISION EQUATIONS [ALGB-01]

Method: Division & Multiplication Properties of Equality & Reciprocal Multipl..

#1: Solve 8x = 136 and prove with [f] [LASTX] [×].
SC-6 Keys: `136 [ENTER] 8 [÷]` **Ans: x = 17**

#2: Solve 12y = 180 and prove with [f] [LASTX] [×].
SC-6 Keys: `180 [ENTER] 12 [÷]` **Ans: y = 15**

#3: Solve m / 6 = 19 and prove with [f] [LASTX] [÷].
SC-6 Keys: `19 [ENTER] 6 [×]` **Ans: m = 114**

#4: Solve k / 7 = 14.5.
SC-6 Keys: `14.5 [ENTER] 7 [×]` **Ans: k = 101.5**

#5: Solve 4.5w = 63.
SC-6 Keys: `63 [ENTER] 4.5 [÷]` **Ans: w = 14**

#6: Solve 3/4 x = 18.
SC-6 Keys: `18 [ENTER] [.] 3 [.] 4 [÷]` **Ans: x = 24**

#7: Solve 2/5 n = 16 using BOTH [÷] and [f] [1/x] [×].
SC-6 Keys: `16 [ENTER] [.] 2 [.] 5 [÷]` **Ans: n = 40**

#8: Solve 5/6 p = 2/3.
SC-6 Keys: `[.] 2 [.] 3 [ENTER] [.] 5 [.] 6 [÷]` **Ans: p = 4/5 (0.8)**

#9*: A 3D printer prints at 1.5 cm/min. How long (t) to print a 27 cm tower (1.5t = 27)? Solve & prove!
SC-6 Keys: `27 [ENTER] 1.5 [÷]` **Ans: 18 minutes (Check: 18 × 1...**

#10*: 3/5 of the students in a club (3/5 s = 27) are 6th graders. How many total students s are in the club?
SC-6 Keys: `27 [ENTER] [.] 3 [.] 5 [÷]` **Ans: 45 students (Way 2: 27 ÷ 3..**

SEC 7.4: WRITING, GRAPHING & SOLVING INEQUALITIES ([F] [>], [G] [<], [F] [≥], [G] [≤]) [..

Method: Writing, Testing & Graphing Inequalities on a Number Line (Open Circl..

#1: Solve x + 8 > 21. Tell whether the circle at 13 is Open or Solid, and arrow direction.
SC-6 Keys: `21 [ENTER] 8 [-]` **Ans: x > 13 (Open circle, arrow..**

#2: Solve y - 9 ≤ 15. Tell whether the circle at 24 is Open or Solid, and arrow direction.
SC-6 Keys: `15 [ENTER] 9 [+]` **Ans: y ≤ 24 (Solid dot, arrow L..**

#3: Solve 6n ≥ 78. Tell whether the circle at 13 is Open or Solid, and arrow direction.
SC-6 Keys: `78 [ENTER] 6 [÷]` **Ans: n ≥ 13 (Solid dot, arrow R..**

#4: Solve m / 4 < 9. Tell whether the circle at 36 is Open or Solid, and arrow direction.
SC-6 Keys: `9 [ENTER] 4 [×]` **Ans: m < 36 (Open circle, arrow..**

#5: Translate: 'You must be at least 48 inches tall (h) to ride.' Write the inequality!
SC-6 Keys: `'At least' = ≥` **Ans: h ≥ 48**

#6: Translate: 'An elevator holds at most 2,000 pounds (w).' Write the inequality!
SC-6 Keys: `'At most' = ≤` **Ans: w ≤ 2000**

#7: Which numbers from {4, 5, 6} are solutions to 4n + 3 > 23? Test on your SC-6!
SC-6 Keys: `4(5)+3=23 (False); 4(6)+3=27 (True)` **Ans: Only 6 (At n=5, 23 > 23 is..**

#8: Which numbers from {4, 5, 6} are solutions to 4x - 3 ≥ 17?
SC-6 Keys: `4(5)-3=17 ≥ 17 (1=True)` **Ans: 5 and 6**

#9*: You have $50 to buy $8 books (8b ≤ 50). Solve with [g] [÷R] to find the max whole books b you can buy!
SC-6 Keys: `50 [ENTER] 8 [g] [÷R]` **Ans: At most 6 books ($2 left o..**

#10*: A gamer needs at least 500 points (p + 340 ≥ 500). Solve for p AND test p = 160 with [f] [≥]!
SC-6 Keys: `500 [ENTER] 340 [-]` **Ans: p ≥ 160 (340+160=500 ≥ 500..**

SEC 7.5: TWO-VARIABLE RELATIONSHIPS (Y = KX AND Y = X + B) — TABLES & GRAPHS [COUNT-01]

Method: Independent (Input x) & Dependent (Output y) Variables, Function Tabl..

#1: Table has (1, 8), (2, 16), (3, 24). Write the equation and find y when x = 10.
SC-6 Keys: `16 [ENTER] 2 [÷] -> 8; 8×10` **Ans: y = 8x; y = 80**

#2: Table has (1, 9), (2, 10), (5, 13). Write the equation and find y when x = 20.
SC-6 Keys: `9 [ENTER] 1 [-] -> +8; 20+8` **Ans: y = x + 8; y = 28**

#3: Table has (2, 14), (5, 35), (8, 56). Write the equation and find y when x = 12.
SC-6 Keys: `14 [ENTER] 2 [÷] -> 7; 7×12` **Ans: y = 7x; y = 84**

#4: Table has (10, 6), (15, 11), (22, 18). Write the equation and find y when x = 30.
SC-6 Keys: `30 [ENTER] 4 [-]` **Ans: y = x - 4; y = 26**

#5: Use 6 [g] [REP] [+] to generate the y-values of y = 6x for x = 1, 2, 3, 4.
SC-6 Keys: `6 [g] [REP] [+] [+] [+]` **Ans: (1,6), (2,12), (3,18), (4,..**

#6: In d = 65t (distance d in miles after t hours), which variable is Independent and which is Dependent?
SC-6 Keys: `d depends on t` **Ans: Independent: t; Dependent: d**

#7: Using d = 65t, how far does the car travel in 4.5 hours?
SC-6 Keys: `65 [ENTER] 4.5 [×]` **Ans: 292.5 miles**

#8: Does the graph of y = 4x pass through the origin (0, 0)? What about y = x + 4?
SC-6 Keys: `Test x = 0 in both!` **Ans: y = 4x hits (0,0); y = x +..**

#9*: You earn p = 12.50h. How many hours h must you work to earn $100? Solve by division AND ratio doubling!
SC-6 Keys: `100 [ENTER] 12.50 [÷]` **Ans: 8 hours (Way 2: 1h=12.5 ->..**

#10*: A gym charges a $25 signup fee plus $15 per month (m). Write the equation for total cost c and find c for 12..
SC-6 Keys: `15 [ENTER] 12 [×] 25 [+]` **Ans: c = 15m + 25; $205 for 12..**

Unit 8: Surface Area & Volume of 3D Solids — Complete Key

Big Ideas Math Grade 6 — Chapter 8 • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Gridfinity 3x1 Storage Boxes

Target Misconception: Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).

Hands-On Block Remediation: Have students touch the 3 pairs of faces on their physical Gridfinity box while logging each pair into [Σ+] (n=1, 2, 3), use [f] [PRISM] ([f][9]) to multiply 3 stack floors (l, w, h) at once for volume, and unlock [f] [CIRC], [g] [CONE], and [g] [HYP].

Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!



SEC 8.1: 3D SOLIDS, NETS & SURFACE AREA OF RECTANGULAR PRISMS ([F] [RECT] + [Σ+]) [COUNT-01]

Method: Parallelogram Cut-and-Slide (A = bh) & Triangle as Half of a Parallel..

#1: Find the surface area of a cube with edge length s = 4 cm (6s²).
SC-6 Keys: `4 [x²] 6 [×]` **Ans: 96 cm²**

#2: Find the surface area of a cube with edge length s = 9 in.
SC-6 Keys: `9 [x²] 6 [×]` **Ans: 486 in²**

#3: Find the surface area of a 6 cm by 4 cm by 3 cm rectangular prism using the 3-pair [Σ+] ledger.
SC-6 Keys: `2(24)+2(18)+2(12)` **Ans: 108 cm²**

#4: Find the surface area of a 10 in by 5 in by 2 in rectangular box.
SC-6 Keys: `2(50)+2(20)+2(10)` **Ans: 160 in²**

#5: Find the surface area of a prism with l = 12 ft, w = 8 ft, and h = 5 ft.
SC-6 Keys: `2(96)+2(60)+2(40)` **Ans: 392 ft²**

#6: Find the surface area of a prism with decimal edges: 5 m by 4 m by 2.5 m.
SC-6 Keys: `2(20)+2(12.5)+2(10)` **Ans: 85 m²**

#7: A cube has a total surface area of 150 cm². What is the area of 1 face, and what is the edge length s?
SC-6 Keys: `150 [ENTER] 6 [÷] [f][√x]` **Ans: 1 face = 25 cm²; s = 5 cm**

#8: Spot the Glitch: For a 5×4×3 box, Leo added 20 + 15 + 12 = 47. What did he forget to do?
SC-6 Keys: `47 [ENTER] 2 [×]` **Ans: 94 (He only found 3 faces;..**

#9*: An open-top 3D-printed bin is 10 cm long, 8 cm wide, and 5 cm tall (NO lid!). Find its outer surface area 2 w..
SC-6 Keys: `80 + 2(50) + 2(40)` **Ans: 260 cm² (Way 2: Full box 3..**

#10*: You wrap a gift box (15 in × 10 in × 4 in). If you have 550 in² of wrapping paper, is it enough?
SC-6 Keys: `2(150)+2(60)+2(40)` **Ans: 500 in² needed -> Yes, 50..**

SEC 8.2: SURFACE AREA OF TRIANGULAR PRISMS & PYRAMIDS ([F] [RECT] + [G] [TRI] + [Σ+]) [COUNT-01]

Method: Trapezoid Area Formula A = ½h(b1 + b2) & Decomposing Composite Figure..

#1: Square Pyramid: Base is 6 cm × 6 cm; 4 triangles each have b = 6 cm, slant height = 5 cm. Find SA.
SC-6 Keys: `6 [x²] 6 [ENTER] 5 [g][TRI] 4 [×] [+]` **Ans: 96 cm²**

#2: Square Pyramid: Base is 8 in × 8 in; 4 triangles each have b = 8 in, slant height = 10 in. Find SA.
SC-6 Keys: `8 [x²] 8 [ENTER] 10 [g][TRI] 4 [×] [+]` **Ans: 224 in²**

#3: Square Pyramid: Base is 12 ft × 12 ft; 4 triangles each have b = 12 ft, slant height = 9 ft. Find SA.
SC-6 Keys: `12 [x²] 12 [ENTER] 9 [g][TRI] 4 [×] [+]` **Ans: 360 ft²**

#4: Regular Tetrahedron: Made of 4 identical equilateral triangles (b = 6 cm, h = 5.2 cm). Find total SA.
SC-6 Keys: `6 [ENTER] 5.2 [g][TRI] 4 [×]` **Ans: 62.4 cm²**

#5: Triangular Prism: 2 triangle ends (b=4, h=3) + 3 rectangles of length 10 (widths 3, 4, 5). Find SA.
SC-6 Keys: `4 [ENTER] 3 [g][TRI] 2 [×] (12) + 12×..` **Ans: 132**

#6: Triangular Prism Tent: 2 triangle ends (b=8 ft, h=3 ft) + 3 rectangles of length 7 ft (widths 5, 5, 8 ft).
SC-6 Keys: `8 [ENTER] 3 [g][TRI] 2 [×] (24) + 18×..` **Ans: 150 ft²**

#7: A regular tetrahedron has 4 identical triangles with b = 8 cm and h = 6.93 cm. Find its surface area.
SC-6 Keys: `8 [ENTER] 6.93 [g][TRI] 4 [×]` **Ans: 110.88 cm²**

#8: How many faces (triangles and rectangles) does a Triangular Prism have vs. a Square Pyramid?
SC-6 Keys: `Count net faces` **Ans: Prism: 2 tris + 3 rects; P..**

#9*: Camping Tent (Triangular Prism): 2 ends (b=6 ft, h=4 ft), floor (6×8 ft), and 2 roof sides (5×8 ft each). Find t..
SC-6 Keys: `24 + 48 + 80` **Ans: 152 ft² (Way 2: 24 + (6+5+..**

#10*: A glass pyramid skylight has NO bottom floor—only 4 triangular glass panes (b = 9 ft, slant h = 6 ft). Find gl..
SC-6 Keys: `9 [ENTER] 6 [g][TRI] 4 [×]` **Ans: 108 ft² (Way 2: 27 + 27 +..**

SEC 8.3: VOLUME OF RECTANGULAR PRISMS WITH WHOLE & FRACTIONAL EDGES ([F] [PRISM]) [COUNT..

Method: Unfolding 3D Solids into 2D Nets — Rectangular Prisms SA = 2(lw + lh..

#1: Use [f] [PRISM] to find the volume of a box measuring 10 cm by 6 cm by 4 cm.
SC-6 Keys: `10 [ENTER] 6 [ENTER] 4 [f][PRISM]` **Ans: 240 cm³**

#2: Find the volume of a cube with edge length 5 in using [g] [x³] or [f] [PRISM].
SC-6 Keys: `5 [g][x³]` **Ans: 125 in³**

#3: Find the volume of a crate measuring 15 ft by 8 ft by 6 ft.
SC-6 Keys: `15 [ENTER] 8 [ENTER] 6 [f][PRISM]` **Ans: 720 ft³**

#4: A prism has Base Area B = 32 m² and height h = 7 m. Find its volume.
SC-6 Keys: `32 [ENTER] 7 [×]` **Ans: 224 m³**

#5: Fractional Edges: Find the volume of a fish tank 3 1/2 ft long, 1 1/2 ft wide, and 2 ft tall.
SC-6 Keys: `3.5 [ENTER] 1.5 [ENTER] 2 [f][PRISM]` **Ans: 10 1/2 ft³ (10.5)**

#6: Fractional Edges: Find the volume of a block 4 1/2 in by 3 in by 2 in.
SC-6 Keys: `4.5 [ENTER] 3 [ENTER] 2 [f][PRISM]` **Ans: 27 in³**

#7: Missing Height: A box has Volume = 270 in³, length = 9 in, and width = 5 in. Find height h.
SC-6 Keys: `270 [ENTER] 9 [ENTER] 5 [×] [÷]` **Ans: 6 in**

#8: How many 1/2-inch mini-cubes fit inside 1 cubic inch? (Compute 2³!)
SC-6 Keys: `2 [g][x³]` **Ans: 8 mini-cubes per 1 in³**

#9*: A planter has Base Area 45 in² (9×5) and Volume 360 in³. Find depth h by division AND verify with [f] [PRIS..
SC-6 Keys: `360 [ENTER] 45 [÷]` **Ans: 8 in (Check: 9 [ENTER] 5 [..**

#10*: How many 1/2-inch cubes fit inside a box with volume 27 in³? (Multiply 27 by 2³ = 8!)
SC-6 Keys: `27 [ENTER] 2 [g][x³] [×]` **Ans: 216 half-inch cubes**

SEC 8.4: CIRCLES, CONES & RIGHT-TRIANGLE HYPOTENUSE ([▨], [F] [CIRC], [G] [CONE], [G] [H..

Method: 3D Layer Stacking Volume (V = B · h = l · w · h) & Fractional Unit-Cu..

#1: Find the area of a circle with radius r = 5 cm using [f] [CIRC].
SC-6 Keys: `5 [f][CIRC]` **Ans: 78.54 cm²**

#2: Find the area of a circle with radius r = 10 in using [f] [CIRC].
SC-6 Keys: `10 [f][CIRC]` **Ans: 314.16 in²**

#3: Find the hypotenuse of a right triangle with legs a = 3 and b = 4 using [g] [HYP].
SC-6 Keys: `3 [ENTER] 4 [g][HYP]` **Ans: 5**

#4: Find the hypotenuse of a right triangle with legs a = 6 ft and b = 8 ft using [g] [HYP].
SC-6 Keys: `6 [ENTER] 8 [g][HYP]` **Ans: 10 ft**

#5: Find the hypotenuse of a skate ramp with base 12 ft and back height 5 ft using [g] [HYP].
SC-6 Keys: `12 [ENTER] 5 [g][HYP]` **Ans: 13 ft**

#6: A circle has diameter d = 16 m. Divide by 2 to find radius r, then press [f] [CIRC]!
SC-6 Keys: `16 [ENTER] 2 [÷] [f][CIRC]` **Ans: 201.06 m²**

#7: Find the volume of a cone with radius r = 3 cm and height h = 12 cm using [g] [CONE].
SC-6 Keys: `3 [ENTER] 12 [g][CONE]` **Ans: 113.10 cm³**

#8: Find the volume of a cone with radius r = 6 in and height h = 5 in using [g] [CONE].
SC-6 Keys: `6 [ENTER] 5 [g][CONE]` **Ans: 188.50 in³**

#9*: Find the volume of a cone (r = 3 cm, h = 10 cm) both with [g] [CONE] AND via 3 [f] [CIRC] 10 [×] 3 [÷]!
SC-6 Keys: `3 [ENTER] 10 [g][CONE]` **Ans: 94.25 cm³ (Both ways match..**

#10*: TV screens are measured along the diagonal. A screen is 32 in wide and 24 in tall. Find its diagonal size 2..
SC-6 Keys: `32 [ENTER] 24 [g][HYP]` **Ans: 40-inch screen (Way 2: √(1..**

Unit 9: Statistical Measures & Data Displays — Complete Key

Big Ideas Math Grade 6 — Chapters 9 & 10 (Statistical Measures and Data Displays) • Manipulative Kits: Counting & Array Tiles (COUNT-01) + Probability Dice Studio (PROB-01)

Target Misconception: Students forget to sort data before finding Median/Quartiles, lose their place adding long lists for Mean, and dread Mean Absolute Deviation (MAD) because doing it by hand takes 20 minutes of subtraction.

Hands-On Block Remediation: Show that Mean is physically leveling unequal cube towers to the same height. Then show that entering a list ONCE with [Σ+] unlocks Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), Range ([RNG]), Quartiles ([f] [1Q], [g] [3Q], [IQR]), and MAD ([g] [MAD]) in 1 click each!



Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!

SEC 9.1: DOT PLOTS, FREQUENCY & MEAN AS 'FAIR SHARE LEVELING' ([Σ+], [G] [x-bar]) [COUNT-01]

Method: Statistical Questions (Variability), Dot Plots & Fair-Share Leveling for the Mean (x-bar = Σx / n) | E..

#1: Level the towers 2, 4, 6, 8. Find the sum ([f] [Σx]) and Mean ([g] [x-bar]).
SC-6 Keys: [g][CLΣ] 2 [Σ+] 4 [Σ+] 6 [Σ+] 8 [Σ+] [g][x-bar] **Ans: Sum = 20, Mean = 5**

#2: Find the sum and Mean of: 12, 15, 18, 11, 14.
SC-6 Keys: Load 5 nums into [Σ+] -> [g][x-bar] **Ans: Sum = 70, Mean = 14**

#3: Find the Mean of your 5 gaming scores: 18, 24, 15, 27, 21.
SC-6 Keys: Load into [Σ+] -> [g][x-bar] **Ans: Mean = 21**

#4: Find the Mean of these 4 decimal weights (g): 4.2, 3.8, 4.5, 3.5.
SC-6 Keys: Load into [Σ+] -> [g][x-bar] **Ans: Mean = 4.0 g**

#5: Typo Rescue: Enter 10 [Σ+] 20 [Σ+] 35 [Σ+], then remove 35 with 35 [f] [Σ-] and add 25 [Σ+]. Find [g] [x-bar]!
SC-6 Keys: ... 35 [f][Σ-] 25 [Σ+] [g][x-bar] **Ans: Sum = 55, Mean = 18.33**

#6: Frequency Shortcut ([FRQ]): If 10 appears 4 times and 20 appears 2 times, find the Mean of all 6 numbers!
SC-6 Keys: 10 [ENTER] 4 [FRQ] 20 [ENTER] 2 [FRQ].. **Ans: 13.33 (80 ÷ 6)**

#7: Find the Mean daily high temperature (°F): 68, 72, 75, 71, 64, 70.
SC-6 Keys: Load 6 temps into [Σ+] -> [g][x-bar] **Ans: 70°F**

#8: If 5 towers have a Mean height of 12 blocks, how many total blocks are in all 5 towers combined?
SC-6 Keys: 5 [ENTER] 12 [×] **Ans: 60 total blocks**

#9*: You need a Mean of 90 across 4 quizzes (360 pts). Your first 3 are 85, 92, 88. Find Quiz 4 score 2 ways!
SC-6 Keys: 360 [ENTER] 265 [-] **Ans: 95 (Way 2: (-5)+(+2)+(-2)=..**

#10*: 6 players have a Mean score of 15 pts (90 pts). A 7th player scores 29 pts. What is the new Mean of all 7?
SC-6 Keys: 90 [ENTER] 29 [+] 7 [÷] **Ans: 17 points**

SEC 9.2: MEDIAN, MODE & RANGE — ONE DATA ENTRY, FIVE INSTANT ANSWERS! [PROB-01]

Method: Ordering Data Least-to-Greatest for Median (Middle), Mode (Most Frequ..

#1: Load 4, 9, 2, 7, 8 into [Σ+]. Find the Median ([f] [MED]) and Range ([RNG]).
SC-6 Keys: Load 5 nums -> [f][MED], [RNG] **Ans: Median = 7, Range = 7 (9 -..**

#2: Load 12, 15, 12, 18, 20, 13 into [Σ+]. Find the Median, Mode ([g] [MODE]), and Range.
SC-6 Keys: Load 6 nums -> [f][MED], [g][MODE], [.. **Ans: Median = 14, Mode = 12, Ra..**

#3: Find the Mean, Median, Mode, and Range of: 5, 8, 8, 10, 14.
SC-6 Keys: Load 5 nums into [Σ+] **Ans: Mean = 9, Med = 8, Mode =..**

#4: Find the Minimum ([f] [MIN]), Maximum ([g] [MAX]), and Range ([RNG]) of: 23, 45, 19, 38, 51.
SC-6 Keys: Load 5 nums -> [f][MIN], [g][MAX], [R.. **Ans: Min = 19, Max = 51, Range..**

#5: Even Count Median: Find the Median of the 6 test scores 70, 80, 84, 90, 92, 98.
SC-6 Keys: Load 6 nums -> [f][MED] **Ans: 87 ((84+90)÷2)**

#6: Outlier Check: Earnings are $15, $18, $15, $20, $17, $90. Find Mean and Median. Which represents a typic..
SC-6 Keys: Load 6 nums -> [g][x-bar], [f][MED] **Ans: Mean=$29.17, Med=$17.50 (M..**

#7: What happens to the Mean and Median of {10, 12, 14, 16, 18} if 18 changes to 100?
SC-6 Keys: Test both on [Σ+]! **Ans: Mean jumps 14 -> 30.8; Med..**

#8: Mystery Range: The minimum score on a quiz is 64 and the Range is 31. What was the maximum score?
SC-6 Keys: 64 [ENTER] 31 [+] **Ans: 95**

#9*: Find Median and Range of {14, 8, 21, 11, 16} both by sorting tiles by hand AND with [f] [MED] & [RNG]!
SC-6 Keys: Sorted: 8, 11, 14, 16, 21 **Ans: Median = 14, Range = 13 (2..**

#10*: Home prices on a block (in $1,000s): 210, 220, 215, 225, 980. Find Mean and Median—which should a buy..
SC-6 Keys: Load 5 nums -> [g][x-bar], [f][MED] **Ans: Mean=$370k, Med=$220k (Tru..**

SEC 9.3: QUARTILES, INTERQUARTILE RANGE ([IQR]) & BOX-AND-WHISKER PLOTS [FRAC-01]

Method: Measures of Variation — Range (Max - Min), Quartiles (Q1, Q3) & Inter..

#1: Load 10, 20, 30, 40, 50 into [Σ+]. Record Min, Q1, Med, Q3, Max, and [IQR].
SC-6 Keys: Load 5 nums -> [f][1Q], [g][3Q], [IQR] **Ans: Min=10, Q1=15, Med=30, Q3=..**

#2: Load 4, 6, 8, 10, 12, 14, 16 into [Σ+]. Find Q1, Median, Q3, and [IQR].
SC-6 Keys: Load 7 nums -> [f][1Q], [g][3Q], [IQR] **Ans: Q1=6, Med=10, Q3=14, IQR=8**

#3: If a box plot has Q1 = 24 and Q3 = 58, what is the Interquartile Range (IQR)?
SC-6 Keys: 58 [ENTER] 24 [-] **Ans: 34**

#4: What percent of data in a Box Plot falls above Q1 (the 25th percentile)?
SC-6 Keys: 3 quarters = 3 × 25% **Ans: 75%**

#5: Find the 5-Number Summary and [IQR] of the 9 scores: 68, 72, 75, 78, 82, 85, 88, 92, 96.
SC-6 Keys: Load 9 nums into [Σ+] **Ans: 68, 73.5, 82, 90, 96; IQR..**

#6: Find Q1, Median, Q3, and [IQR] for the 8 speeds: 42, 45, 48, 50, 52, 55, 58, 60.
SC-6 Keys: Load 8 nums into [Σ+] **Ans: Q1=46.5, Med=51, Q3=56.5,..**

#7: Consistency Check: Player A has IQR = 4 pts and Player B has IQR = 19 pts. Who is more consistent?
SC-6 Keys: Compare box widths **Ans: Player A (smaller IQR = ti..**

#8: Outlier Rule (1.5 × IQR): If Q1 = 40, Q3 = 60 (IQR = 20), any score above Q3 + 1.5(IQR) is an outlier. Find t..
SC-6 Keys: 60 [ENTER] 1.5 [ENTER] 20 [×] [+] **Ans: 90 (Any score > 90 is an o..**

#9*: For {12, 15, 18, 22, 25, 28, 34}, find [IQR] both with the [IQR] key AND by subtracting [g] [3Q] - [f] [1Q]!
SC-6 Keys: Load 7 nums -> [IQR] **Ans: IQR = 13 (28 - 15 = 13)**

#10*: 200 students take a test. The box plot has Min=50, Q1=70, Med=80, Q3=90, Max=100. How many students..
SC-6 Keys: 200 [ENTER] 50 [f][%] **Ans: 100 students (50% of 200)**

SEC 9.4: MEAN ABSOLUTE DEVIATION ([G] [MAD]) WITHOUT THE 20-MINUTE NIGHTMARE [COUNT-01]

Method: Mean Absolute Deviation (MAD) — Average Distance of Data Values from..

#1: Load 2, 4, 6 into [Σ+]. Find the Mean ([g] [x-bar]) and MAD ([g] [MAD]).
SC-6 Keys: 2 [Σ+] 4 [Σ+] 6 [Σ+] [g][MAD] **Ans: Mean = 4, MAD = 1.33 (4/3)**

#2: Load 10, 12, 14, 16, 18 into [Σ+]. Find the Mean and MAD.
SC-6 Keys: Load 5 nums -> [g][x-bar], [g][MAD] **Ans: Mean = 14, MAD = 2.4**

#3: Load 5, 8, 8, 10, 14 into [Σ+]. Find the Mean and MAD.
SC-6 Keys: Load 5 nums -> [g][x-bar], [g][MAD] **Ans: Mean = 9, MAD = 2.4**

#4: What is the MAD of 7, 7, 7, 7, 7 (5 identical towers)?
SC-6 Keys: All distances from 7 are 0! **Ans: Mean = 7, MAD = 0**

#5: Find the Mean and MAD of: 20, 24, 28, 32, 36.
SC-6 Keys: Load 5 nums -> [g][x-bar], [g][MAD] **Ans: Mean = 28, MAD = 4.8**

#6: Find the Mean and MAD of the 6 sprint times: 12.4, 13.0, 12.6, 13.2, 12.8, 14.0.
SC-6 Keys: Load 6 decimals -> [g][x-bar], [g][MAD] **Ans: Mean = 13.0, MAD = 0.40**

#7: Find the Mean and MAD of quiz scores: 82, 85, 88, 90, 95.
SC-6 Keys: Load 5 nums -> [g][x-bar], [g][MAD] **Ans: Mean = 88, MAD = 3.6**

#8: Team A has Mean = 50, MAD = 2.1. Team B has Mean = 50, MAD = 8.4. Which team's scores are more spr..
SC-6 Keys: Compare MADs **Ans: Team B (larger MAD = more..**

#9*: For {6, 10, 11, 13}, find MAD both by averaging |x - 10| distances AND with 1 click of [g] [MAD]!
SC-6 Keys: Mean=10; dists 4+0+1+3=8 ÷ 4 **Ans: Mean = 10, MAD = 2.0**

#10*: Full Chapter 9 Boss Sweep: For {5, 8, 8, 10, 14}, record [g][x-bar], [f][MED], [g][MODE], [RNG], [IQR], and [..
SC-6 Keys: 1 entry -> 6 stats! **Ans: x-bar=9, Med=8, Mode=8, Rn..**

Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability — Complete..

Big Ideas Math Grade 7 Bridge — Chapters 1–3 & Chapter 7 • Manipulative Kits: Ten-Frame Zero-Pair Counters (BASE-10) + Algebra Stage (ALGB-01) + Probability Studio (PROB-01)

Target Misconception: Students guess signs when adding/subtracting negatives, get -25 instead of +25 for (-5)² on algebraic calculators, undo ×3 before +7 in 3x + 7 = 28, and confuse Permutations (order matters) with Combinations (order doesn't).

Hands-On Block Remediation: Use Yellow/Red counters to show removing 4 red debt tokens (10 - (-4)) equals +14. Notice how RPN 5 [+/-] [x²] squares the entire register (-5)² = +25 without parentheses! Unwrap 2-step equations in reverse (untie ribbon ±b first, open box ×a second), and finish with [DICE], [f] [nCr], [g] [nPr], and [f] [?].

Optional QUEST-01 Scoreboard: 1 '+10' block = Solved 1 Way; stack a 2nd '+10' block (+20) = Solved 2 Ways (Blocks + Textbook/SC-6)!



SEC 10.1: ADDING & SUBTRACTING SIGNED NUMBERS — WHY 'MINUS A NEGATIVE' ADDS! [BASE-10]

Method: Zero-Pair Chip Cancellation (1 Positive + 1 Negative = 0) & Add-the-O..

#1: Solve side-by-side: (a) -15 + 9 and (b) -15 + (-9).
SC-6 Keys: `15 [+/-] [ENTER] 9 [+] / 9 [+/-] [+]`
Ans: (a) -6; (b) -24

#2: Solve: 14 + (-23).
SC-6 Keys: `14 [ENTER] 23 [+/-] [+]`
Ans: -9

#3: Taking Away a Penalty: Solve (a) 12 - (-8) and (b) -12 - (-8).
SC-6 Keys: `12 [ENTER] 8 [+/-] [-] / 12 [+/-]...`
Ans: (a) 20; (b) -4

#4: Solve: -9 - 16. (Start at -9 and go 16 steps further negative!)
SC-6 Keys: `9 [+/-] [ENTER] 16 [-]`
Ans: -25

#5: Solve with decimals: -4.8 + (-3.5).
SC-6 Keys: `4.8 [+/-] [ENTER] 3.5 [+/-] [+]`
Ans: -8.3

#6: Solve with decimals: 6.4 - 11.2.
SC-6 Keys: `6.4 [ENTER] 11.2 [-]`
Ans: -4.8

#7: Solve with decimals: -7.5 - (-10.2).
SC-6 Keys: `7.5 [+/-] [ENTER] 10.2 [+/-] [-]`
Ans: +2.7

#8: Solve with signed fractions: -3/4 - (-5/8).
SC-6 Keys: `[.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8..`
Ans: -1/8

#9*: Temperature in Fairbanks is -18°F at dawn and rises to +7°F by noon. Find the change (7 - (-18)) 2 ways!
SC-6 Keys: `7 [ENTER] 18 [+/-] [-]`
Ans: +25°F (Way 2: 18 steps to..

#10*: Evaluate -2 1/2 + 4 1/4 - (-1 1/8) on the SC-6 and check by converting to eighths!
SC-6 Keys: `2.5 [+/-] [ENTER] 4.25 [+] 1.125 [+]`
Ans: 2 7/8 (2.875)

SEC 10.2: MULTIPLYING & DIVIDING SIGNED RATIONALS — THE LIGHT-SWITCH FLIP RULE [BASE-10]

Method: Sign Rules for Multiplying & Dividing Rational Numbers (Same Signs =..

#1: Compute: (a) 7 × (-8) and (b) (-7) × (-8).
SC-6 Keys: `7 [ENTER] 8 [+/-] [×] / 7 [+/-]...`
Ans: (a) -56; (b) +56

#2: Compute: (a) -45 ÷ 9 and (b) -45 ÷ (-9).
SC-6 Keys: `45 [+/-] [ENTER] 9 [÷] / 9 [+/-] [÷]`
Ans: (a) -5; (b) +5

#3: Compute (-5)² using 5 [+/-] [x²]. Why is it +25?
SC-6 Keys: `5 [+/-] [x²]`
Ans: +25 (2 negatives = +)

#4: Compute (-2)³ using 2 [+/-] [g] [x³]. Why is a negative cubed still negative?
SC-6 Keys: `2 [+/-] [g][x³]`
Ans: -8 (3 negatives = odd = -)

#5: Multiply signed decimals: (-3.5) × (-2.4).
SC-6 Keys: `3.5 [+/-] [ENTER] 2.4 [+/-] [×]`
Ans: +8.4

#6: Divide signed decimals: -14.4 ÷ 1.2.
SC-6 Keys: `14.4 [+/-] [ENTER] 1.2 [÷]`
Ans: -12

#7: Multiply signed fractions: (-2/3) × (9/10).
SC-6 Keys: `[.] 2 [.] 3 [+/-] [ENTER] [.] 9 [.] 1..`
Ans: -3/5 (-0.6)

#8: Divide signed fractions: -5/6 ÷ (-5/12).
SC-6 Keys: `[.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 1..`
Ans: +2

#9*: Triple Product: Evaluate (-4) × (-5) × (-3) both by counting negative signs AND on your SC-6!
SC-6 Keys: `4 [+/-] [ENTER] 5 [+/-] [×] 3 [+/-] [×]`**Ans: -60 (3 negatives = odd ->..**

#10*: A submarine dives 360 ft in 8 minutes (-45 ft/min). Confirm the total change in 8 min and the change per mi..
SC-6 Keys: `360 [+/-] [ENTER] 8 [÷]`
Ans: -45 ft/min (Check: -45 × 8..

SEC 10.3: SOLVING TWO-STEP EQUATIONS (AX + B = C) — UNWRAPPING IN REVERSE! [ALGB-01]

Method: Solving Two-Step Equations (ax + b = c) by Undoing Addition/Subtracti..

#1: Solve 4x + 9 = 45. Write the Step 1 value (4x = ?) and final x.
SC-6 Keys: `45 [ENTER] 9 [-] 4 [÷]`
Ans: 4x = 36 -> x = 9

#2: Solve 5x - 12 = 38. (Undo -12 with [+] first, then ÷ 5!)
SC-6 Keys: `38 [ENTER] 12 [+] 5 [÷]`
Ans: 5x = 50 -> x = 10

#3: Solve 6y + 11 = 59.
SC-6 Keys: `59 [ENTER] 11 [-] 6 [÷]`
Ans: 6y = 48 -> y = 8

#4: Solve 8n - 15 = 49.
SC-6 Keys: `49 [ENTER] 15 [+] 8 [÷]`
Ans: 8n = 64 -> n = 8

#5: Solve m / 4 + 6 = 15. (Subtract 6 first, then multiply by 4!)
SC-6 Keys: `15 [ENTER] 6 [-] 4 [×]`
Ans: m/4 = 9 -> m = 36

#6: Solve k / 3 - 5 = 7.
SC-6 Keys: `7 [ENTER] 5 [+] 3 [×]`
Ans: k/3 = 12 -> k = 36

#7: Solve -3x + 14 = 2.
SC-6 Keys: `2 [ENTER] 14 [-] 3 [+/-] [÷]`
Ans: -3x = -12 -> x = 4

#8: Solve 2.5w + 8 = 23.
SC-6 Keys: `23 [ENTER] 8 [-] 2.5 [÷]`
Ans: 2.5w = 15 -> w = 6

#9*: A repair tech charges $35 flat plus $25/hr (25h + 35 = 160). Find hours h by unwrapping AND plug-in check!
SC-6 Keys: `160 [ENTER] 35 [-] 25 [÷]`
Ans: h = 5 hours (Check: 25(5)+..

#10*: You save $15/week toward a $120 skateboard and already have $45 (15w + 45 = 120). How many weeks w?
SC-6 Keys: `120 [ENTER] 45 [-] 15 [÷]`
Ans: w = 5 weeks (Check: 15(5)+..

SEC 10.4: PROBABILITY, COMBINATORICS & GRADUATION QUIZ SPRINT ([DICE], [X!], [F] [NCR],..

Method: Circle Circumference (C = πd = 2πr) & Area (A = πr²), Theoretical Probability & Counting (n!, nPr, n..

#1: Roll a fair 6-sided die. Find P(rolling a 5) as a fraction, and P(rolling an even number) in simplest form.
SC-6 Keys: `1/6; and 3/6 -> [g][SIMP]`
Ans: 1/6; and 1/2 (50%)

#2: A 10-sector spinner has 4 Blue, 3 Gold, 3 Red. Find P(Blue) as a simplified fraction, decimal, and percent.
SC-6 Keys: `[.] 4 [.] 10 [g][SIMP] [FDISP]`
Ans: 2/5, 0.4, 40%

#3: Using the same 10-sector spinner, find P(NOT Red) as a fraction and percent.
SC-6 Keys: `(10 - 3) / 10 = 7/10`
Ans: 7/10 (70%)

#4: In how many ways can 4 students line up at the door? Compute 4! using [g] [x!].
SC-6 Keys: `4 [g][x!]`
Ans: 24 ways

#5: In how many ways can you arrange 6 books on a shelf? Compute 6! using [g] [x!].
SC-6 Keys: `6 [g][x!]`
Ans: 720 ways

#6: How many ways can you choose a 3-person team from 8 players (order does NOT matter: [f] [nCr])?
SC-6 Keys: `8 [ENTER] 3 [f][nCr]`
Ans: 56 teams

#7: How many ways can 1st, 2nd, and 3rd place be awarded among 8 finalists (order DOES matter: [g] [nPr])?
SC-6 Keys: `8 [ENTER] 3 [g][nPr]`
Ans: 336 podiums

#8: Two Independent Events: You flip a coin (1/2) AND roll a 6-sided die (P(>4) = 2/6 = 1/3). Multiply to find P(H..
SC-6 Keys: `[.] 1 [.] 2 [ENTER] [.] 1 [.] 3 [×]`
Ans: 1/6

#9*: From 10 toppings, compare choosing 3 toppings ([f] [nCr]) vs. 3 ordered medals ([g] [nPr]). How are they rel..
SC-6 Keys: `10 [ENTER] 3 [f][nCr] / [g][nPr]`
Ans: 120 vs 720 (120 × 3! = 720)

#10*: GRADUATION BOSS: Compute 5! both with [g] [x!] AND 5 [ENTER] 5 [g] [nPr], then complete a 5-streak on..
SC-6 Keys: `5 [g][x!]; then [f][?]`
Ans: 120 ways (Course Complete!)